

Spectral extremes under exact cycle conditioning

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Abstract

Let P_n be the matrix of a random permutation of n symbols and let $M_n = \log \max_{|z|=1} |\det(I - zP_n)|$. Cook and Zeitouni proved that $M_n / \log n$ converges in probability to a constant x_0 for a uniform permutation. We show that the $\sqrt{\log n}$ fluctuations of M_n are carried entirely by the number of cycles K_n . Write $\lambda(s) = \log\{\Gamma(1+s)/\Gamma(1+s/2)^2\}$, let s_κ be the minimizer of $(1 + \kappa\lambda(s))/s$ on $(0, \infty)$, and put $v(\kappa) = \kappa\lambda'(s_\kappa)$ and $a_\theta = \lambda(s_\theta)/s_\theta$. Under the Ewens measure with any fixed parameter $\theta > 0$,

$$M_n = v(\theta) \log n + a_\theta(K_n - \theta \log n) + O_{\mathbb{P}}(\log \log n),$$

so that the standardized pair (K_n, M_n) converges jointly to (G, G) with G standard normal: the maximum and the cycle count are asymptotically perfectly aligned. This is deduced from a statement about the exact conditional law, which does not depend on θ : for every compact $[\kappa_-, \kappa_+] \subset (0, \infty)$ there is a finite C with

$$\sup_{\kappa_- - \log n \leq k \leq \kappa_+ \log n} \mathbb{P}(|M_n - v(k/\log n) \log n| > C \log \log n \mid K_n = k) \longrightarrow 0,$$

uniformly over exact and possibly atypical cycle counts. The proof keeps the size and the cycle count simultaneously in a two-variable coefficient extraction. Cycles longer than $n/(\log n)^4$ are reserved as an analytic factor whose coefficients are flat under every size shift produced by the shorter cycles; positivity then converts a scalar coefficient asymptotic into a relative comparison of the entire path-constrained measure, with an error that does not degrade with the number of constraints or with the rarity of the event. The constrained lower bound comes from pointwise saddle estimates for killed convolutions along a dyadic chain of endpoint boxes.

1 Introduction

1.1 The maximum of a random permutation characteristic polynomial

Let σ_n be a random permutation of $\{1, \dots, n\}$, let P_n be its permutation matrix, and let $C_j = C_j(\sigma_n)$ be its number of j -cycles. Write $e(t) = \exp(2\pi it)$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ with normalized Lebesgue measure. The characteristic polynomial factors along cycles,

$$F_n(z) = \det(I - zP_n) = \prod_{j \geq 1} (1 - z^j)^{C_j},$$

so that on the unit circle

$$\log |F_n(e(t))| = \sum_{j \geq 1} C_j \log |1 - e(jt)|. \quad (1.1)$$

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We study

$$M_n = \log \|F_n\|_\infty = \max_{t \in \mathbb{T}} \log |F_n(e(t))|,$$

which is also the logarithm of the maximum modulus of $\det(zI - P_n)$.

The field (1.1) is of log-correlated type. For a uniform permutation the counts C_j are asymptotically independent Poisson variables of mean $1/j$, so (1.1) is a sum over $\asymp \log n$ effective dyadic scales of frequencies, its variance is of order $\log n$, and the covariance of two points decays like the logarithm of their distance. Its increments are, however, far from Gaussian: the summand $\log |1 - e(jt)|$ is bounded above by $\log 2$ and has an exponential left tail. The extremal behaviour of M_n is therefore governed by a genuine Legendre transform and not by a Gaussian rate. The relevant cumulant generating function is that of $\log |1 - e(U)|$ for a uniform $U \in \mathbb{T}$,

$$A(s) = \int_{\mathbb{T}} |1 - e(t)|^s dt = \frac{\Gamma(1+s)}{\Gamma(1+s/2)^2}, \quad \lambda(s) = \log A(s), \quad s > -1. \quad (1.2)$$

For complex z with $\operatorname{Re} z > -1$ we write $A(z)$ for the same integral, with $|1 - e(t)|^z = \exp(z \log |1 - e(t)|)$ off the roots and value zero at the roots; it is analytic and equals the same gamma ratio.

Cook and Zeitouni [3, Theorem 1.2] proved that for a uniform permutation $M_n/\log n$ converges in probability to a constant x_0 , characterized by $\lambda^*(x_0) = 1$ for the Legendre transform λ^* of λ . Their proof combines arithmetic separation of frequencies, a division into small and large arcs, two-point estimates and a truncated second moment. Those mechanisms reappear below, at the level of the constrained kernels; what is new here is the way the exact combinatorial constraints are carried through them.

1.2 Why the cycle count should be conditioned on

The randomness in (1.1) has two layers of very different character. The arithmetic layer is the choice of the frequencies j that occur; it produces the log-correlated structure and all the difficulty in locating the maximum. The other layer is the total number of cycles

$$K_n = \sum_{j \geq 1} C_j,$$

a single global statistic that merely counts how many summands (1.1) has. Under the Ewens measure with parameter $\theta > 0$ one has $K_n = \theta \log n + O_{\mathbb{P}}(\sqrt{\log n})$, and for the maximum this fluctuation is not negligible. The extremal speed depends on the number of summands, so a displacement of order $\sqrt{\log n}$ in K_n moves M_n by the same order; and $\sqrt{\log n}$ dominates every correction of size $\log \log n$. The leading fluctuation of M_n is therefore inherited from the count layer, not from the arithmetic one.

This suggests separating the two layers, and in the Ewens family the separation is exact. The cycle counts satisfy

$$\mathbb{P}_{\theta,n}(C_j = c_j) = h_n(\theta)^{-1} \prod_{j \geq 1} \frac{(\theta/j)^{c_j}}{c_j!}, \quad \sum_j j c_j = n,$$

where

$$h_n(u) = [z^n](1-z)^{-u}, \quad a_{n,k} = [u^k]h_n(u), \quad h_0(u) = 1, \quad (1.3)$$

and coefficients with an impossible index are zero. Conditioning on $K_n = k$ cancels the factor θ^k and leaves

$$\mathbb{P}_{n,k}(C_j = c_j) = a_{n,k}^{-1} \prod_j \frac{j^{-c_j}}{c_j!}, \quad \sum_j j c_j = n, \quad \sum_j c_j = k. \quad (1.4)$$

This law does not depend on θ . It is a microcanonical law, with the size and the number of cycles both fixed exactly, and it is the canonical object that remains once the count layer has been removed.

Working with (1.4) is strictly more demanding than working with the Ewens law itself. The two are related by the mixture identity

$$\mathbb{P}_{\theta,n}(A_n) = \sum_k \mathbb{P}_{\theta,n}(K_n = k) \mathbb{P}_{n,k}(A_n), \quad (1.5)$$

and convergence of its left-hand side to zero gives no uniform bound on the conditional probabilities on the right: at the mode $\mathbb{P}_{\theta,n}(K_n = k)$ is only of order $(\log n)^{-1/2}$, and it is exponentially smaller for counts away from $\theta \log n$. A statement about the exact conditional law has to be proved directly, and every estimate that carries the size constraint has to keep a *relative* error throughout. Producing such an estimate for a rare path event, *after* all of the constraints defining that event have been imposed, is the technical heart of this paper.

Exact cycle conditioning also arises on its own. In [1] the author showed that for two uniformly random perfect matchings on $2n$ labels the profile of alternating components is Ewens(1/2); there the number of components is the natural quantity to fix, and Section 6 records the resulting spectral statement.

1.3 Results

For each $\kappa > 0$ the function $s \mapsto (1 + \kappa \lambda(s))/s$ has a unique minimizer s_κ on $(0, \infty)$, characterized by

$$\kappa \{s_\kappa \lambda'(s_\kappa) - \lambda(s_\kappa)\} = 1. \quad (1.6)$$

Indeed $\lambda''(s) > 0$ is the variance of $\log |1 - e(U)|$ under the exponential tilt of order s , so the left-hand side vanishes at $s = 0$, has strictly positive derivative $\kappa s \lambda''(s)$ for $s > 0$, and tends to infinity by Stirling's formula. Put $v(\kappa) = \kappa \lambda'(s_\kappa)$. Then, with $\kappa = k/\log n$,

$$m_{n,k} := \inf_{s>0} \frac{\log n + k \lambda(s)}{s} = \frac{\log n + k \lambda(s_\kappa)}{s_\kappa} = v(\kappa) \log n. \quad (1.7)$$

Theorem 1.1 (Exact-cycle localization). *Let $0 < \kappa_- < \kappa_+ < \infty$. There is a finite constant C , depending only on this interval, such that*

$$\sup_{\substack{k \in \mathbb{Z} \\ \kappa_- \log n \leq k \leq \kappa_+ \log n}} \mathbb{P}_{n,k}(|M_n - m_{n,k}| > C \log \log n) \longrightarrow 0. \quad (1.8)$$

The constant is chosen before the error tolerance and serves every admissible integer k at once, including counts far from the typical Ewens value. Both the upper and the lower half of (1.8) are proved under the exact constraint; neither is obtained by averaging over K_n .

Theorem 1.1 transfers back to the Ewens family. Fix $\theta > 0$ and put $a_\theta = \lambda(s_\theta)/s_\theta > 0$. Since $K_n/\log n \rightarrow \theta$ in probability, one may apply (1.8) with $\kappa = K_n/\log n$ and expand v to first order at θ ; the quadratic remainder is $O_{\mathbb{P}}(1)$. With probability tending to one and with a fixed constant,

$$|M_n - v(\theta) \log n - a_\theta(K_n - \theta \log n)| \leq C \log \log n, \quad (1.9)$$

and consequently, by the classical central limit theorem for K_n ,

$$\left(\frac{K_n - \theta \log n}{\sqrt{\theta \log n}}, \frac{M_n - v(\theta) \log n}{a_\theta \sqrt{\theta \log n}} \right) \Longrightarrow (G, G), \quad G \sim \mathcal{N}(0, 1). \quad (1.10)$$

This is Corollary 6.1. The extremal speed is set by an environment — by how many cycles the permutation happens to have — and that environment is itself asymptotically Gaussian; the maximum follows it. Statement (1.10) is joint weak convergence, and no assertion about moments or about a limit for the Pearson correlation is made.

1.4 Discussion

The speed function. The map $\kappa \mapsto v(\kappa)$ is strictly increasing and strictly concave: implicit differentiation of (1.6) gives (6.1), namely $v'(\kappa) = \lambda(s_\kappa)/s_\kappa > 0$ and $v''(\kappa) = -\{\kappa^3 s_\kappa^3 \lambda''(s_\kappa)\}^{-1} < 0$. More cycles raise the extremal speed, with diminishing returns. The first derivative is exactly the coefficient a_θ in (1.9), which is why the random centring is linear in K_n at the accuracy considered here. Since $\lambda'(s) < \log 2$ for every $s > 0$ we have $v(\kappa) < \kappa \log 2$, in agreement with the deterministic bound $M_n \leq K_n \log 2$; the two are asymptotically equal as $\kappa \downarrow 0$, where a permutation with few cycles can make all of its factors nearly extremal at once. At $\kappa = 1$ equation (1.6) is the Legendre-transform equation $\lambda^*(v(1)) = 1$, so $v(1) = x_0$ is the Cook–Zeitouni constant and Theorem 1.1 contains their leading order.

The localization window. The width $C \log \log n$ in (1.8) is the natural limit of the present method. For branching random walks the second-order term of the maximum is logarithmic in the number of generations, and the number of effective generations here is of order $\log n$; a correction of size $\log \log n$ is therefore expected in (1.8), and Theorem 1.1 does not resolve it. We identify neither the coefficient of such a correction, nor a limit law for the residual, nor an extremal point process. On the other hand $\log \log n \ll \sqrt{\log n}$, so the window is far below the scale of (1.9) and (1.10); those two statements are sharp at leading order.

Remark 1.2. Since (1.4) does not involve θ , Theorem 1.1 applies verbatim to every law on cycle profiles of the form $\mathbb{P}(c) \propto w(\sum_j c_j) \prod_j j^{-c_j}/c_j!$ with $w \geq 0$: conditioning any such law on its cycle count returns (1.4). Whenever the induced count concentrates on a compact set of values of $k/\log n$, it follows that $|M_n - m_{n,K_n}| \leq C \log \log n$ with probability tending to one for that law. The Ewens family is the case $w(k) = \theta^k$.

1.5 Relation to previous work

Beyond [3], several strands bear on the problem. Hughes, Najnudel, Nikeghbali and Zeindler [4] study moments, multiplicative class functions and linear statistics of characteristic polynomials under generalized Ewens measures, with generating functions in cycle counts. Dang and Zeindler [5] obtain joint central limit theorems at finitely many fixed observation points under arithmetic hypotheses on those points. François [6, Theorem 2.2] proves convergence of generalized-Ewens characteristic polynomials in the topology of local uniform convergence on the open unit disk. Neither a finite collection of observation points nor convergence in the interior controls the maximum on the boundary, which is a supremum over a family of constraints whose size grows with n . The central limit theorem for K_n used in (1.10) is classical; sharper uniform asymptotics for the Ewens count distribution are in [7], and the coefficient asymptotics behind (1.3) belong to the classical theory of Stirling numbers of the first kind.

Random centring of an extremum by an environment has precedents. For branching random walks in a time-inhomogeneous random environment, Mallein and Miloš [8] identify an environment-dependent centre together with a logarithmic correction. The mechanism behind (1.9) is of that type, with the cycle count playing the role of the environment, although the model here is a finite permutation and the proof is self-contained: it uses elementary count couplings, path estimates and second moments rather than an extremal theorem for random environments.

Cook and Gu have announced related work on maxima of Poissonian log-correlated fields [9]. Public abstracts from January 2025, October 2025 and January 2026 describe refined maximal behaviour for related random series or trigonometric polynomials, modelled on a branching random walk in a random time-dependent environment, and the research listing describes the work as in preparation. A complete theorem statement was not available in the

sources consulted on September 10, 2026, so no comparison with a full theorem, and no claim of priority over that work, is made here. We note only that the object treated below is the exact finite permutation law under a precise cycle-count constraint, for which (1.5) shows that an unconditional statement is not sufficient.

The coefficient viewpoint continues that of [1]. The scalar principles behind the reservoir estimate of Section 2 originate in the hybrid method of Flajolet, Fusy, Gourdon, Panario and Pouyanne [2]; the estimate needed here has a cutoff that grows with n and a complex marker, and is proved directly. The contribution of the coefficient step is the uniformity of its relative error after all restrictions of a rare path event have been imposed on a positive measure.

1.6 Outline of the proof

Throughout the proof $L = \log n$, $\ell = \log L$ and $b = \lfloor n/L^4 \rfloor$. The following is the skeleton; the constants are fixed in the order set out in Section 5.

An analytic reservoir. A fixed-observation hybrid estimate cannot control the growing family of path restrictions that a maximum requires. Instead we reserve every cycle longer than b . What remains is a polynomial whose coefficients are positive measures recording the heights $\log |1 - e(jt)|$ at the observation points, and whose degree is $O(Lb) = o(n)$ under the exact count constraint. The reserved factor $R_b(z, u) = \exp(u \sum_{j>b} z^j/j)$ has one algebraic singularity and an entire correction; a uniform Hankel estimate and a saddle point in the marker u show that its coefficients are flat under every size shift the short cycles can produce, with relative error $O(1/\ell)$. Because the comparison is between positive measures, it can be applied *after* all path projections and yields a relative comparison of the whole restricted measure. Neither the number of projections, nor the dimension of the space, nor the mass of the event enters the error. This is Theorem 2.3, and it is what replaces a total-variation approximation whose absolute error would exceed the probability of the event being measured.

A constrained kernel lower bound. After the transfer, the short cycles may be treated as independent harmonic samples in logarithmic blocks. Grouping the blocks dyadically from both ends produces $B \asymp \ell$ coarse groups of widths H_j that vary by a bounded factor. Along a chain of endpoint boxes of width $\sqrt{H_j}$ we bound the killed convolution kernels from below pointwise, using a bridge estimate for the tilted log-sine law (Lemma 3.2). Integrating over the intermediate boxes cancels the density factors $H_j^{-1/2}$ against the box widths, so that the surviving cost is $\sum_j E_j^2$ rather than $B \max_j E_j^2$, where E_j measures the deviation of the realized counts in group j . This is what keeps the total cost polynomial, $p_L \geq L^{-C^*}$ (Lemma 3.4), and it is what limits the final window to $O(\ell)$.

Fourier comparison and a second moment. On arithmetically separated points the tilted block kernels are close, in total variation $O(L^{-J})$ for any fixed J , to independent copies of the reference convolution (Lemma 4.2). Since J may be chosen after C_* , that error is $o(p_L^2)$ and the comparison is valid *relative* to the rare event, not merely in absolute terms. A first and second moment for the constrained functional then give a high point in any prescribed set of angles of measure at least one half; the matching upper bound follows from an integrated moment estimate whose input is an entropy gap $\kappa b_s < 1$ for the block envelope (Lemma 4.3).

Restoring the remaining cycles. The low-frequency cycles are restored on a random but admissible set of angles, using only that the lower bound of the previous step is uniform over such sets. The long cycles are reinserted by a deterministic stability estimate for the supremum norm under multiplication by $\prod(1 - z^{j_v})$ (Lemma 5.1), whose loss is controlled by an exact

conditional factorial-moment identity for $(\sum_{j \leq b} j C_j)(\sum_{l > b} C_l/l)$ (Lemma 5.2). Both operations move the centre by $O(\ell)$ only.

1.7 Organization

Section 2 constructs the reservoir, proves the event-uniform coefficient comparison and identifies the law of the marked block counts. Section 3 proves the pointwise killed-convolution estimates, the regularity of the count environment and the box-chain lower bound. Section 4 supplies the uniform one- and two-point Fourier comparisons. Section 5 fixes the constants, runs the moment computation, restores the low and long cycles and completes the proof of Theorem 1.1. Section 6 derives Corollary 6.1, records the two-matching application and delimits what is and is not asserted.

2 An analytic reservoir and a positive coefficient transfer

Throughout the proof set

$$L = \log n, \quad \ell = \log L, \quad b = \lfloor n/L^4 \rfloor, \quad H_b = \sum_{j=1}^b \frac{1}{j}, \quad T = L - H_b = 4\ell - \gamma + o(1). \quad (2.1)$$

All assertions are for sufficiently large n . Define

$$R_b(z, u) = \exp \left(u \sum_{j>b} \frac{z^j}{j} \right) = (1-z)^{-u} \exp\{-uH_b(z)\}, \quad H_b(z) = \sum_{j=1}^b \frac{z^j}{j}, \quad (2.2)$$

and $r_{N,l}^{(b)} = [z^N u^l] R_b(z, u)$.

2.1 A uniform scalar coefficient estimate

Lemma 2.1 (Reservoir coefficient estimate). *Fix a compact set $\mathcal{U} \subset \mathbb{C}$. Uniformly for $u \in \mathcal{U}$, $b \geq 2$, and $N \geq 4b$,*

$$\begin{aligned} [z^N] R_b(z, u) &= e^{-uH_b} h_N(u) + E_{N,b}(u), \\ |E_{N,b}(u)| &\leq C \frac{b}{N} e^{-\operatorname{Re}(u)H_b} N^{\operatorname{Re}(u)-1} + C b^C e^{-N/(2b)}. \end{aligned} \quad (2.3)$$

The estimate is absolute, including at zeros of $1/\Gamma(u)$.

Proof. Continue (2.2) to the disk of radius $1 + 1/b$, slit along $[1, 1 + 1/b]$, using the branch of $\log(1-z)$ analytic at zero. Put $\rho = 1 + 1/b$. On $|z| = \rho$ the quantity $H_b(z) + \log(1-z)$ is bounded by an absolute constant on either side of the slit. To verify this, for $|\arg z| \leq 2/b$ use $|H_b(z) - H_b| \leq Cb|z-1| \leq C$ and $\log|1-z| = -\log b + O(1)$. Otherwise the identity

$$H_b(z) + \log(1-z) = - \int_0^z \frac{w^b}{1-w} dw$$

on the radial segment gives a bound $C/(b|\arg z|) \leq C$; here $|1 - te^{i\varphi}| \geq c(|1-t| + |\varphi|)$ for $0 \leq t \leq \rho$, $|\varphi| \leq \pi$, after harmless changes of constants. Thus R_b is bounded on the outer circle, uniformly on $\mathcal{U}$. The comparison function $e^{-uH_b}(1-z)^{-u}$ has at worst polynomial size in b there. Their outer-contour contributions to the coefficient are at most $Cb^C \rho^{-N} \leq Cb^C e^{-N/(2b)}$.

Deform Cauchy's contour into the outer circle, both sides of the slit, and an indentation $|z-1| = 1/N$. For $|w| \leq 1/b$,

$$|H_b(1+w) - H_b| \leq Cb|w|, \quad |e^{-u(H_b(1+w)-H_b)} - 1| \leq Cb|w|.$$

On the indentation, $|z|^{-N-1} \leq C$ and its length is $O(N^{-1})$. The contribution of the difference of the two functions is at most $Ce^{-aH_b}N^{a-1}(b/N)$, where $a = \operatorname{Re} u$. On either side of the slit write $z = 1 + t$, $1/N \leq t \leq 1/b$. The remaining contribution is bounded by

$$Ce^{-aH_b}b \int_{1/N}^{1/b} t^{1-a} e^{-Nt/2} dt \leq Ce^{-aH_b}N^{a-1} \frac{b}{N}.$$

The last integral is uniform for a in a compact interval; its lower limit is $1/N$, so it poses no integrability problem when $a \geq 2$. Branch moduli contribute bounded factors on $\mathcal{U}$. This proves (2.3), without moving the indentation to zero.

Uniformity of the contour bounds. Here and below a compact complex marker set is enclosed in a disk $|u| \leq M$, with M fixed. The estimate on the outer circle can be checked without an infinite series outside its disk of convergence. For $z = \rho e^{i\varphi}$ with $|\varphi| > 2/b$, use the radial integral above and $|1 - te^{i\varphi}| \geq c|\varphi|$. Its absolute value is at most

$$\frac{C}{|\varphi|} \int_0^\rho t^b dt = \frac{C\rho^{b+1}}{(b+1)|\varphi|} \leq \frac{C}{b|\varphi|}.$$

For $|\varphi| \leq 2/b$, one has $|z - 1| \asymp b^{-1}$ and $|z - 1| \leq C/b$. The elementary identity $z^j - 1 = (z - 1) \sum_{v=0}^{j-1} z^v$ gives $|H_b(z) - H_b| \leq Cb|z - 1|$. Since $H_b = \log b + O(1)$ and the branch argument of $1 - z$ is bounded, this also bounds $H_b(z) + \log(1 - z)$ in the small-angle sector. These arguments apply to the two boundary values at the slit.

For clarity, the remaining power of N in the bank integral is obtained by the substitution $v = Nt$:

$$b \int_{1/N}^{1/b} t^{1-a} e^{-Nt/2} dt = bN^{a-2} \int_1^{N/b} v^{1-a} e^{-v/2} dv \leq C_M b N^{a-2}, \quad |a| \leq M.$$

The last constant is finite because the lower endpoint is one and the upper tail is exponential, uniformly in a . On the indentation, $|(1 - z)^{-u}| \leq C_M N^a$, the analytic correction is $O_M(b/N)$, and the arc length is $O(1/N)$. All terms are therefore absolute estimates of the same difference of analytic functions. In particular none was obtained by division by $h_N(u)$ or by $1/\Gamma(u)$; their possible zeros cause no exception. □

Lemma 2.2 (Marker extraction). *Uniformly for $k/\log n$ in a positive compact interval,*

$$a_{n,k} = \frac{L^k}{nk! \Gamma(k/L)} \{1 + O(L^{-1})\}. \quad (2.4)$$

Let b be as in (2.1), $0 \leq d \leq C_0 Lb$, $N = n - d$, and $T_N = \log N - H_b$. Uniformly when l/T_N belongs to a positive compact interval,

$$r_{N,l}^{(b)} = \frac{T_N^l}{N! \Gamma(l/T_N)} \left\{ 1 + O\left(T_N^{-1} + \sqrt{T_N} b/N\right) \right\}. \quad (2.5)$$

Consequently, if l/T belongs to a fixed positive compact interval,

$$\sup_{0 \leq d \leq C_0 Lb} \left| \frac{r_{n-d,l}^{(b)}}{r_{n,l}^{(b)}} - 1 \right| \leq \varepsilon_n, \quad \varepsilon_n = O(\ell^{-1} + L^{-3}) = o(1). \quad (2.6)$$

Proof. For bounded u , the gamma-ratio formula, interpreted analytically at nonpositive integers, gives

$$h_N(u) = N^{u-1} \left\{ \frac{1}{\Gamma(u)} + O(N^{-1}) \right\},$$

with an absolute uniform error of modulus $CN^{\operatorname{Re} u - 2}$. For an entire function g , $x \rightarrow \infty$, and j/x in a positive compact interval, the elementary marker saddle is

$$[u^j]e^{xu}g(u) = \frac{x^j}{j!} \{g(j/x) + O(x^{-1})\}. \quad (2.7)$$

For completeness, write $g(u) = \sum g_m u^m$ and divide the left-hand side by $x^j/j!$. It becomes $\sum_{m \leq j} g_m(j)_m/x^m$. On a disk of radius greater than twice the largest j/x , Cauchy's coefficient bound and $|1 - (j)_m/j^m| \leq m(m-1)/(2j)$ give the error $O(j^{-1})$. The omitted tail is exponentially small. This proves (2.7); it also holds for a function analytic on such a fixed disk.

Apply it with $g = 1/\Gamma$ and $x = L$. The coefficient of the gamma-ratio error on $|u| = k/L$ is at most $Cn^{-2}(k/L)^{-k}e^k$, which is smaller than the main term by $O(\sqrt{L}/n)$. This proves (2.4). For the reservoir, Lemma 2.1 gives the main function $N^{-1}e^{T_N u}/\Gamma(u)$. On $|u| = l/T_N$, Cauchy's bound for its error, compared with Stirling's estimate for $T_N^l/l!$, is $O(\sqrt{T_N}b/N)$ relatively. The outer-contour error is negligible: $N/b \asymp L^4$ whereas $\log N \asymp L$, so even a fixed polynomial in N, b times $e^{-N/(2b)}$ tends to zero faster than any negative power of L . The main-function saddle gives the additional $O(T_N^{-1})$.

Finally $T_N - T = \log(N/n) = O(d/n)$, and the logarithmic derivative in N of $T_N^l/(N\Gamma(l/T_N))$ is $O(N^{-1})$ uniformly in the indicated range. Equivalently, take the ratio of the explicit main terms. Together with (2.5), this proves (2.6).

The two normalizations in the marker argument. To detail the power-series estimate, put $t = j/x$ and choose a fixed radius $R_0 > 2 \sup t$. If $|g_m| \leq M_0 R_0^{-m}$, then

$$\left| \sum_{m=0}^j g_m \frac{(j)_m}{x^m} - g(t) \right| \leq \frac{M_0}{2j} \sum_{m=0}^j m(m-1)(t/R_0)^m + M_0 \sum_{m>j} (t/R_0)^m = O(x^{-1}).$$

The bound for $(j)_m$ follows from $1 - \prod_{v=0}^{m-1} (1 - v/j) \leq \sum_{v=0}^{m-1} v/j$. This calculation is valid for complex coefficients as well. For $g = 1/\Gamma$ its value at the positive saddle is bounded away from zero on the prescribed interval. Thus the absolute $O(x^{-1})$ error in the braces becomes a relative error there.

In the reservoir application, let $t = l/T_N$. The Cauchy error coming from the first term of (2.3) is at most $C(b/N^2)t^{-l}e^{tT_N}$. The positive main coefficient has size $N^{-1}(T_N^l/l!)g(t)$. Their ratio is at most

$$C \frac{b}{N} \frac{e^l l!}{l!} \leq C \frac{b}{N} \sqrt{l} = O(\sqrt{T_N}b/N).$$

For flatness, keep the integer l and the cutoff b fixed while the real size variable N moves from $n - d$ to n . If $F(N) = T_N^l/(N\Gamma(l/T_N))$, and $\psi = \Gamma'/\Gamma$ on the positive real axis, then

$$\frac{d}{dN} \log F(N) = \frac{1}{N} \left\{ -1 + \frac{l}{T_N} + \frac{l}{T_N^2} \psi(l/T_N) \right\}.$$

The braces are uniformly bounded. Also $d/n = O(L^{-3})$ and $T_N \asymp T \asymp \ell$. Integrating the derivative bounds the ratio of main terms by $1 + O(d/n)$; the two relative saddle errors add $O(T^{-1} + \sqrt{T}b/n)$. This proves the stated ε_n uniformly over every allowed size shift with a single constant. □

2.2 One transfer for the entire constrained kernel

For finite positive measures, write $\mu \leq \nu$ if the inequality holds on every measurable set.

Theorem 2.3 (Positive coefficient transfer). *Let l/T stay in a positive compact interval and let $\mathbf{B}(z) = \sum_{d=0}^D z^d \nu_d$, where each ν_d is a finite positive measure on any measurable space and $D \leq C_0 Lb$. Then*

$$(1 - \varepsilon_n) r_{n,l}^{(b)} \mathbf{B}(1) \leq [z^n] \{[u^l] R_b(z, u) \mathbf{B}(z)\} \leq (1 + \varepsilon_n) r_{n,l}^{(b)} \mathbf{B}(1). \quad (2.8)$$

In particular the bound holds after any collection of positive path projections or weights. Its error does not depend on the dimension of the space, the number of projections, or the mass of the event being measured. If $\mathbf{B}(1)$ has positive total mass, normalizing both sides to probability measures changes the multiplicative error only by $1 + O(\varepsilon_n)$.

Proof. The middle measure is exactly $\sum_{d=0}^D r_{n-d,l}^{(b)} \nu_d$. Apply (2.6) term by term. The last assertion follows by applying the same inequalities to the whole space and dividing. No norm bound for a product of projection operators is needed.

Normalization and simultaneous restrictions. Here is the exact division used in the last assertion. Write $\nu = \sum_d \nu_d$, $r = r_{n,l}^{(b)}$, and $\mu = \sum_d r_{n-d,l}^{(b)} \nu_d$. For large n , $r > 0$ and $0 \leq \varepsilon_n < 1$. If $0 < \nu(\Omega) < \infty$, the whole-space estimate gives $0 < \mu(\Omega) < \infty$. For every measurable E ,

$$\frac{1 - \varepsilon_n}{1 + \varepsilon_n} \frac{\nu(E)}{\nu(\Omega)} \leq \frac{\mu(E)}{\mu(\Omega)} \leq \frac{1 + \varepsilon_n}{1 - \varepsilon_n} \frac{\nu(E)}{\nu(\Omega)}.$$

If $\nu(E) = 0$, both the middle and the reference event probabilities are zero, so no division by an event probability is needed. More generally a nonnegative measurable weight w can be included by replacing $\nu_d(E)$ with $\int_E w d\nu_d$, provided these weighted measures are finite. A conjunction of path inequalities is exactly such a weight, namely its indicator. All the inequalities are imposed on the stored increment vector before forming this sum.

For the actual short-cycle application, fix the counts q_i and let $q = \sum_i q_i$, $l = k - q$. The product of the unnormalized harmonic sample masses has total mass $\prod_i H_i^{q_i}/q_i!$. Grouping it by its total size gives the measures ν_d above. Every possible size is at most $qb \leq kb$. Repeated samples and the factorial $q_i!$ give precisely the cycle weights in (1.4); on normalization this is the usual multinomial expansion of the harmonic sampling law. Thus the reference event E can specify the entire multiblock path, its endpoint, and restrictions on several observation points at once. The one reservoir factor depends only on d and l , irrespective of how that event was described. □

Here is its concrete use. Partition $\{1, \dots, b\}$ into blocks $I_0, \dots, I_m$, and put $H_i = \sum_{j \in I_i} 1/j$. For observation points $t_1, \dots, t_d$ let $\mathbf{f}(j) = (\log |1 - e(jt_v)|)_{v=1}^d$. A root is recorded as $-\infty$ in the corresponding coordinate. Heights lie in $(\mathbb{R} \cup \{-\infty\})^d$, with the usual absorbing convention for addition; a positive exponential tilt assigns zero mass to such an atom. After selecting counts q_i , a height-marked block is

$$\mathbf{B}_{i,q_i}(z) = \frac{1}{q_i!} \left(\sum_{j \in I_i} \frac{z^j}{j} \delta_{\mathbf{f}(j)} \right)^{*q_i}. \quad (2.9)$$

Convolution adds heights and size; before the final coefficient is taken, one may retain the whole increment vector and impose all cumulative-height inequalities. This gives positive measures with size degree at most $(\sum q_i)b \leq kb$. Evaluation at $z = 1$, after normalization, is exactly q_i

independent harmonic samples in each block, with mass $1/(jH_i)$. Thus, conditionally on all q_i and on $l = k - \sum q_i$ long cycles, (2.8) compares the *entire* actual short-cycle configuration with these product harmonic measures, with relative error $1 + O(\varepsilon_n)$. The original total-size constraint has been integrated out by its analytic coefficient, not replaced by a total-variation approximation whose error is larger than the rare event.

2.3 The exact-cycle distribution of the block counts

Proposition 2.4 (Multinomial count reservoir). *Let $k/L = \kappa \in [\kappa_-, \kappa_+]$ and let $(Q_0^*, \dots, Q_m^*, L_*)$ be multinomial with k trials and probabilities*

$$H_0/L, \dots, H_m/L, T/L. \quad (2.10)$$

They sum to one. Under (1.4), let Q_i count cycles in I_i and L_ count cycles larger than b . Uniformly on*

$$\mathcal{G} = \{|l - \kappa T| \leq T^{3/4}\}, \quad (2.11)$$

the actual probability of each count vector equals the multinomial probability times $1 + o(1)$, with a uniform error. Both distributions assign probability tending to one to $\mathcal{G}$. These assertions hold for arbitrarily many blocks.

Proof. Sum the exact configuration weights at fixed (q_i, l) . All short configurations have size at most $kb \leq \kappa_+ Lb$. By Theorem 2.3, their total weight is

$$\{1 + O(\varepsilon_n)\} r_{n,l}^{(b)} \prod_i \frac{H_i^{q_i}}{q_i!}.$$

Divide by (2.4) and apply (2.5). The ratio to the multinomial mass is

$$\frac{\Gamma(\kappa)}{\Gamma(l/T)} \{1 + O(T^{-1} + \varepsilon_n)\} = 1 + o(1)$$

uniformly on (2.11). A binomial Chernoff bound gives $\mathbb{P}(|L_*^* - \kappa T| > T^{3/4}) \leq 2e^{-c\sqrt{T}}$ uniformly. Summing the uniform relative estimate on this set proves that the actual mass there also tends to one. No assertion about the mass ratio outside this set is necessary. $\square$

3 Pointwise killed-convolution estimates

All constants in this and the following sections can be chosen uniformly for $\kappa \in [\kappa_-, \kappa_+]$. Thus $s = s_\kappa$ ranges over a fixed compact subset of $(0, \infty)$.

3.1 The Mellin kernel and a bridge density lower bound

Let $V = \log |1 - e(U)|$ for uniform $U \in \mathbb{T}$. A change of variables gives

$$f_V(x) = \frac{e^x}{\pi \sqrt{1 - e^{2x/4}}} \mathbf{1}_{\{x < \log 2\}}. \quad (3.1)$$

For $\beta > 0$, define the positive tilted kernel

$$f_\beta(x) = e^{\beta x - \lambda(\beta)} f_V(x), \quad \mu_\beta = \lambda'(\beta). \quad (3.2)$$

Its convolution transform is $A(\beta + it)/A(\beta)$.

Lemma 3.1 (Uniform convolution bounds). *For β in a fixed compact subset of $(0, \infty)$ and all sufficiently large q , let $g_{\beta,q}$ be the continuous version of the centered q -fold convolution density. Then*

$$\sup_x g_{\beta,q}(x) \leq Cq^{-1/2}, \quad g_{\beta,q}(0) \geq cq^{-1/2}.$$

If Y_j have density $f_\beta(\cdot + \mu_\beta)$ independently, then

$$\mathbb{P}\left(\max_{j \leq q} \left| \sum_{v \leq j} Y_v \right| > w\right) \leq 2e^{-cw^2/q}, \quad 1 \leq w \leq cq. \quad (3.3)$$

Proof. The centered log moment generating function is at most Ct^2 for small real t , uniformly in β . The exponential supermartingale, stopped at its first crossing before time q , gives $\mathbb{P}(\max S_j > w) \leq \exp(-tw + Cqt^2)$. Choose t proportional to w/q , and repeat with $-S_j$, proving (3.3).

For $q \geq 3$, the convolution integral is continuous: f_β belongs to $L^{3/2}(\mathbb{R})$, so Young's inequality and continuity of translations give boundedness and continuity of its third and subsequent convolutions. For the density bounds, the centered characteristic function has modulus at most e^{-ct^2} near zero, uniformly on the tilt interval. On each fixed annulus its modulus is at most $\rho < 1$: equality would force e^{itV} to be constant, impossible for the positive density (3.1) on an interval. The gamma ratio and Stirling's formula on vertical strips give the bound $C(1 + |t|)^{-1/2}$ at infinity. Split the Fourier inversion integral into these three regions. The absolute integral is at most $Cq^{-1/2}$. At zero, putting $t = u/\sqrt{q}$ in the first region gives $\sqrt{q}g_{\beta,q}(0) \rightarrow (2\pi\lambda''(\beta))^{-1/2}$ uniformly; the other regions are exponentially small. This proves both bounds. $\square$

For a path of q centered increments write S_j for its partial sums. The density of S_q at zero restricted to $\max_{j \leq q} |S_j| \leq W\sqrt{q}$ is well defined by convolution and Lebesgue integration. All restricted densities below use the iterated convolution integral with the relevant increment density, substituting the prescribed total minus the preceding increments for the last increment.

Lemma 3.2 (A pointwise bridge kernel). *There are fixed $W, c > 0$ such that this restricted density is at least $cq^{-1/2}$, uniformly for the same tilts and all sufficiently large q . Consequently, if $d/q = \lambda'(\beta)$ and β is in the same compact tilt interval, the f_s convolution density at total increment d , restricted to*

$$\left| \sum_{v \leq j} x_v - \frac{j}{q}d \right| \leq W\sqrt{q} \quad (1 \leq j \leq q),$$

is at least

$$cq^{-1/2} \exp\{q[\lambda(\beta) - \lambda(s)] - (\beta - s)d\}. \quad (3.4)$$

In particular, if $|d - q\lambda'(s)|/q = o(1)$, the last exponential is at least $\exp\{-C(d - q\lambda'(s))^2/q\}$.

Proof. Split q into $q_1 = \lfloor q/2 \rfloor$ and $q_2 = q - q_1$. The density at zero contributed by paths having $\max_{j \leq q_1} |S_j| > W\sqrt{q}$ is at most

$$\sup_x g_{\beta,q_2}(x) \mathbb{P}(\max_{j \leq q_1} |S_j| > W\sqrt{q}) \leq Cq^{-1/2}e^{-cW^2}.$$

For a path with total zero, deviations during its second half are negative reversed partial sums of that half. Reversing the increments gives the same bound for a violation in the second half. Subtract the two bounds from $g_{\beta,q}(0) \geq c_0q^{-1/2}$ and choose W large. These are statements about densities, obtained by integrating the unused half-convolution; no conditioning on a null event is used without a density.

On the hyperplane of total increment d , the density ratio of the f_s product to the f_β product is the constant exponential in (3.4). The bridge tube is unchanged by that ratio. Finally λ'' is bounded above and below on the compact tilt interval. The inverse function theorem gives $|\beta - s| \leq C|d - q\lambda'(s)|/q$, and Taylor expansion of the exponent gives the final assertion.

The pointwise integral and its change of tilt. The definition used in this proof can be written explicitly. Given increments $x_1, \dots, x_{q-1}$, set $x_q = d - \sum_{v < q} x_v$ and $S_j = \sum_{v \leq j} x_v$. For a measurable path restriction $E \subset \mathbb{R}^q$, put

$$K_{\beta,q}^E(d) = \int_{\mathbb{R}^{q-1}} \mathbf{1}_E(x_1, \dots, x_q) \prod_{v=1}^q f_{\beta}(x_v) dx_1 \cdots dx_{q-1}.$$

This specifies a value at every d . With no restriction it agrees with the continuous convolution density for $q \geq 3$, by the convolution identities and the regularity established in Lemma 3.1. Integrating in d gives the probability of E under the product increment law: the substitution from the last increment to d is a translation with Jacobian one. The same identity with an endpoint indicator gives disintegration over any measurable endpoint set.

To apply the half-path argument, center the increments by μ_{β} . On the hyperplane of centered total zero, the contribution of a first-half event E_1 is the integral of its first-half product density times $g_{\beta,q_2}(-S_{q_1})$. It is at most $\sup g_{\beta,q_2} \mathbb{P}(E_1)$. For the second half use the reversal $(x_1, \dots, x_q) \mapsto (x_q, \dots, x_1)$. On the free $q-1$ coordinates its absolute Jacobian is one; its product density is unchanged. Moreover a reversed partial sum of length $q-j$ is $-S_j$ when the total is zero. A violation in either half is therefore bounded by one of two identical maximal-inequality estimates. Once W is fixed, their sum is at most $c_0/(2\sqrt{q})$ by increasing W , and then increasing the lower threshold for q so that the allowed range in (3.3) contains $W\sqrt{q}$. This gives a strictly positive bound at the specified endpoint itself.

On the same parametrized hyperplane the change of tilt is exactly

$$\prod_{v=1}^q f_s(x_v) = \exp\{q[\lambda(\beta) - \lambda(s)] - (\beta - s)d\} \prod_{v=1}^q f_{\beta}(x_v).$$

It remains true when one of the factors is zero. For the near-mean assertion first enlarge the compact interval of s slightly within $(-1, \infty)$. On that larger interval let $0 < v_- \leq \lambda'' \leq v_+$. For a fixed small enough $\delta > 0$, $|d/q - \lambda'(s)| \leq \delta$ has a unique solution $\lambda'(\beta) = d/q$ in the enlarged interval, and $|\beta - s| \leq v_-^{-1}|d/q - \lambda'(s)|$. Taylor's formula in the form

$$\lambda(\beta) - \lambda(s) - (\beta - s)\lambda'(\beta) = - \int_s^{\beta} (t - s)\lambda''(t) dt \geq -\frac{v_+}{2}(\beta - s)^2$$

now proves the asserted quadratic penalty, also when $\beta < s$. The enlarged interval is essential when s lies at one endpoint of its original compact range. □

This pointwise estimate is stronger than a lower bound for one endpoint window. It will be integrated over a chain of endpoint boxes. The integration is what prevents payment of a narrow-window factor at every scale.

3.2 Fine scales and a regular count environment

Take a fixed A_0 sufficiently large and fixed constants $g_*, r_* > 0$ to be chosen later. Define

$$h = A_0\ell, \quad r = r_*\ell, \quad a_* = \log b, \quad m = \left\lceil \frac{a_* - r}{h} \right\rceil, \quad \omega = \frac{a_* - r}{m}, \quad a_i = r + i\omega. \quad (3.5)$$

Then $h/2 \leq \omega \leq h$ for large n . Let I_0 contain $j \leq e^r$ and I_i contain $e^{a_{i-1}} < j \leq e^{a_i}$, $1 \leq i \leq m$, with integer endpoints interpreted by these inequalities. They partition $\{1, \dots, b\}$. For the middle counts q_i put

$$S_i = \sum_{j=1}^i q_j, \quad Q = S_m.$$

Their harmonic block masses satisfy

$$H_i = \omega + O(e^{-a_{i-1}}), \quad \sum_{i=1}^m |H_i - \omega| = O(e^{-r}). \quad (3.6)$$

Fix $D_0 > 0$, to be chosen in Lemma 3.4, and put

$$H_*^c = \frac{r^2}{D_0 \log \ell}, \quad v_0 = \lceil H_*^c / \omega \rceil. \quad (3.7)$$

Group the middle blocks dyadically from both ends. Start with $M = m$ remaining blocks and $v = v_0$. While $M \geq 6v$, remove one group of v blocks from each end of the remaining interval, and immediately replace (M, v) by $(M - 2v, 2v)$. At termination $2v \leq M < 6v$. Divide the remaining blocks into $J_c = \lceil M/(2v) \rceil$ consecutive groups, each containing $\lfloor M/J_c \rfloor$ or $\lceil M/J_c \rceil$ blocks, placing the larger groups first. Order all groups from left to right. Set $i_0 = 0$, and let i_j be the last fine index of group j , so that $i_B = m$. Write

$$H_j = a_{i_j} - a_{i_{j-1}}, \quad Q_j = S_{i_j} - S_{i_{j-1}}, \quad 1 \leq j \leq B.$$

For large n this deterministic construction gives

$$c\ell \leq B \leq C\ell, \quad H_j \geq H_*^c, \quad H_1, H_B \in [H_*^c, 2H_*^c], \quad 1/5 \leq H_{j+1}/H_j \leq 5. \quad (3.8)$$

The bounds follow by summing a geometric progression and noting that $m/v_0 \asymp L \log \ell / \ell^2$ with fixed-constant factors. Define

$$E_j = 1 + H_j^{-1/2} \max_{i_{j-1} \leq i \leq i_j} |S_i - S_{i_{j-1}} - \kappa(a_i - a_{i_{j-1}})|. \quad (3.9)$$

For fixed $\eta > 0, K_E, C_E$, call the environment regular if

$$\frac{\kappa}{2}\omega \leq q_i \leq (\kappa + \eta)\omega \quad (1 \leq i \leq m), \quad \max_j E_j \leq K_E \sqrt{\log \ell}, \quad \sum_j E_j^2 \leq C_E B. \quad (3.10)$$

All constants here can be selected uniformly over the compact κ interval.

Lemma 3.3 (Regularity under exact cycle conditioning). *For each fixed $\eta > 0$, constants A_0, K_E, C_E can be chosen, independently of fixed r_*, D_0 , such that under $\mathbb{P}_{n,k}$, with probability tending to one uniformly in k , the environment is regular,*

$$Q_0 \leq Cr, \quad |L_* - \kappa T| \leq T^{3/4}.$$

In particular $L_ \leq C\ell$. The threshold for n may depend on the subsequently fixed r_*, D_0 .*

Proof. By Proposition 2.4, it suffices to prove these assertions for the multinomial vector (2.10). Each q_i is binomial with mean κH_i . A Chernoff bound and (3.6), followed by a union bound over at most L blocks, establish the first part of (3.10) when A_0 is large. The same bound gives $Q_0 \leq Cr$ with probability tending to one. The assertion about L_* was already proved.

Here are details for the multiscale energy; conditioning a fixed number of trials must not be ignored. First form independent Poisson category counts with the same means kp_i . On a coarse group, the centered successive counts are independent-increment martingales. Exponential martingales give

$$\mathbb{P}(E_j > x) \leq Ce^{-cx^2} \quad (2 \leq x \leq c\sqrt{H_j}),$$

uniformly, with the negligible deterministic discrepancies in (3.6) absorbed. The union bound over $B = O(\ell)$ yields the desired maximum bound for a sufficiently large constant. Doob's fourth-moment inequality and $\mathbb{E}(\text{Poi}(\mu) - \mu)^4 = \mu + 3\mu^2$ give $\sup_j \mathbb{E}E_j^4 \leq C$. Different coarse groups

are independent. Hence Chebyshev's inequality shows $\sum E_j^2 \leq C_E B$ with failure probability $O(B^{-1})$ when C_E is sufficiently large.

To pass this elementary count estimate to a fixed number of trials, take an independent $N \sim \text{Poi}(k)$ and one sequence of iid category labels of probabilities (p_i) . The first N labels give the independent Poisson counts, and the first k give the multinomial counts. Conditional on $D = |N - k|$, the differing labels are D iid labels. If D_j is their number in coarse group j , then

$$\mathbb{E} \left[\sum_j \frac{D_j^2}{H_j} \middle| D \right] \leq C \left(\frac{DB}{L} + \frac{D^2}{L} \right).$$

Indeed that category has probability $H_j/L + O(e^{-r}/L)$; use its binomial second moment and (3.8). Since $\mathbb{E}D \leq \sqrt{k}$ and $\mathbb{E}D^2 = k$, the last sum is $O_{\mathbb{P}}(1)$. At every prefix of a coarse group the count change is at most D_j . Thus the squared Euclidean distance between the two vectors of E_j 's is $O_{\mathbb{P}}(1)$. Enlarging K_E, C_E transfers their maximum and energy bounds, since $\sqrt{\log \ell} \rightarrow \infty$ and $B \rightarrow \infty$. This use of Poisson variables is only a proof of a multinomial count inequality. There is no comparison of spectral fields or rare high-point probabilities at this step.

Quantitative effect of fixing the number of trials. The preceding coupling controls all prefixes simultaneously. Write p_j^c for the probability that one category label lies in coarse group j . For large n , $p_j^c \leq CH_j/L$ and $\sum_j p_j^c \leq 1$. Conditional on D , each D_j is binomial with parameters D, p_j^c , even though the D_j are not independent. Hence, without using independence among them,

$$\begin{aligned} \mathbb{E} \left[\sum_j \frac{D_j^2}{H_j} \middle| D \right] &\leq D \sum_j \frac{p_j^c}{H_j} + D^2 \sum_j \frac{(p_j^c)^2}{H_j} \\ &\leq CDB/L + CD^2/L. \end{aligned}$$

Since N is independent of the entire label sequence, the labels between indices $\min(N, k) + 1$ and $\max(N, k)$ remain iid after conditioning on N , hence after conditioning on D for this calculation. Also $\mathbb{E}D^2 = \text{Var } N = k$ and $\mathbb{E}D \leq \sqrt{k}$. The expectation of this energy is bounded uniformly because $B = O(\ell)$ and $k \asymp L$.

Let $\tilde{E}_j$ and E_j be the Poisson and multinomial environments in this one coupling, with the same deterministic centering. The elementary inequality $|\max_i |u_i| - \max_i |v_i|| \leq \max_i |u_i - v_i|$ gives

$$|E_j - \tilde{E}_j| \leq D_j / \sqrt{H_j}, \quad \sum_j E_j^2 \leq 2 \sum_j \tilde{E}_j^2 + 2 \sum_j D_j^2 / H_j.$$

Markov's inequality makes the second sum $o_{\mathbb{P}}(B)$, and its square root $o_{\mathbb{P}}(\sqrt{\log \ell})$. Enlarging the two fixed constants in the Poisson bounds therefore yields both parts of (3.10) for the multinomial vector. No conditioning estimate is divided by $\mathbb{P}(N = k)$, which would lose a factor of order $\sqrt{L}$. □

3.3 Dyadic boxes instead of an assumed barrier theorem

Let $G = g_* \ell$, $s = s_\kappa$, and $\lambda = \lambda(s)$. Define

$$F_i = \frac{a_i + \lambda S_i}{s}, \quad y = \frac{a_* - r + \lambda Q}{s}, \quad B_i = F_i - G. \quad (3.11)$$

In each fine block convolve q_i copies of f_s and one independent noise of modulus at most L^{-10} . Write W_i for the cumulative height through block i . Let

$$\mathcal{A} = \{W_i \leq B_i \ (1 \leq i \leq m), \quad y \leq W_m \leq y + 1\}, \quad p_L = \mathbb{E}_s[e^{-s(W_m - y)} \mathbf{1}_{\mathcal{A}}]. \quad (3.12)$$

This expectation is simply the integral of a product of the positive convolution kernels, with the stated projections.

Lemma 3.4 (Polynomial lower mass for the constrained kernel). *There exist $D_0, C_* < \infty$, depending on the compact κ interval and on A_0, K_E, C_E , with the following property. For all fixed g_*, r_* satisfying*

$$r_* \geq 64s_{\max}g_* + 1, \quad s_{\max} = \max_{\kappa \in [\kappa_-, \kappa_+]} s_{\kappa}, \quad (3.13)$$

and all regular environments, uniformly for large n ,

$$p_L \geq L^{-C_*}. \quad (3.14)$$

The constants D_0, C_ do not depend on the fixed g_*, r_* . The threshold for n is allowed to depend on them.*

Proof. We exhibit a lower bound for a product of killed kernels; no adaptive random-environment measure is an input. Temporarily omit the noises. Choose a large constant B_0 . At a nonterminal coarse endpoint define

$$\gamma_j = 4G + B_0(\sqrt{H_j}E_j + \sqrt{H_{j+1}}E_{j+1}), \quad z_j = F_{i_j} - \gamma_j \quad (1 \leq j < B).$$

At the two ends take

$$z_0 = 0, \quad \gamma_0 = r/s; \quad z_B = y + \frac{1}{2}, \quad \gamma_B = r/s - \frac{1}{2}.$$

Let the intermediate endpoint boxes be $\mathcal{I}_j = [z_j - \sqrt{H_j}, z_j + \sqrt{H_j}]$, $1 \leq j < B$, and let $\mathcal{I}_B = [z_B - 1/10, z_B + 1/10]$. The initial box is the singleton $\{0\}$.

Choose B_0 to dominate the bridge width in Lemma 3.2, the neighboring-width ratio five, and the constants in the next interpolation calculation. Choose D_0 so large that

$$CB_0K_E\sqrt{2/D_0} < 1/(8s_{\max}),$$

where C is a fixed constant large enough for those same bounds. By (3.13), both endpoint gaps then dominate $4G$ plus the required multiple of $E_1\sqrt{H_1}$ or $E_B\sqrt{H_B}$, for large n . Indeed these products are at most $K_Er\sqrt{2/D_0}$, whereas $r/s \geq r/s_{\max}$.

Consider group j and any $x \in \mathcal{I}_{j-1}$, $x' \in \mathcal{I}_j$. Put $d = x' - x$ and $q = Q_j$. The critical identity $s\lambda'(s) - \lambda = 1/\kappa$ gives

$$d - q\lambda'(s) = \frac{H_j - q/\kappa}{s} - (\gamma_j - \gamma_{j-1}) + (e_j - e_{j-1}), \quad (3.15)$$

where $e_j = x' - z_j$ and $e_{j-1} = x - z_{j-1}$. Use $E_0 = E_{B+1} = 1$ when needed. Uniformly over these boxes,

$$|d - q\lambda'(s)| \leq C\sqrt{H_j}U_j, \quad U_j = 1 + E_{j-1} + E_j + E_{j+1} + \frac{r}{\sqrt{H_j}}\mathbf{1}_{\{j=1 \text{ or } j=B\}}. \quad (3.16)$$

For interior groups the $4G$ terms cancel exactly. At the boundary they are bounded by a constant times r . Moreover $q \asymp H_j$, $\max_j U_j/\sqrt{H_j} = O(\log \ell/r) = o(1)$, and

$$\sum_{j=1}^B U_j^2 \leq C \left(B + \sum_j E_j^2 + \frac{r^2}{H_*^c} \right) \leq C\ell. \quad (3.17)$$

The last constant is independent of g_*, r_* , since $r^2/H_*^c = D_0 \log \ell$.

We verify the path restriction before using its density. At a fine endpoint i in this group, set $f = (S_i - S_{i_{j-1}})/q$. The straight line in sample count joining x to x' has gap below F_i equal to

$$(1-f)(\gamma_{j-1} - e_{j-1}) + f(\gamma_j - e_j) + \frac{a_i - a_{i_{j-1}} - fH_j}{s}. \quad (3.18)$$

If $D_i = S_i - S_{i_{j-1}} - \kappa(a_i - a_{i_{j-1}})$, the numerator in its final term is

$$\frac{(a_i - a_{i_{j-1}})D_{i_j} - H_j D_i}{q}.$$

It has modulus at most $CE_j\sqrt{H_j}$. Our choices of gaps, B_0, D_0 make (3.18) exceed $2G + W\sqrt{q}$ at every such endpoint. Hence a bridge lying within $W\sqrt{q}$ of that straight line stays below $F_i - 2G$ at all fine endpoints.

For each fixed pair x, x' , solve the deterministic saddle equation $\lambda'(\beta) = d/q$. It has a solution near s , uniformly by (3.16), and the compactness needed in Lemma 3.2 holds. That lemma gives the following *pointwise* lower bound for the group convolution density, killed at any violated fine barrier:

$$K_j(x, x') \geq cH_j^{-1/2} \exp(-CU_j^2), \quad x \in \mathcal{I}_{j-1}, \quad x' \in \mathcal{I}_j. \quad (3.19)$$

This step chooses a saddle for each density evaluation; it does not assert that the random increments follow a preassigned adaptive law.

Multiply (3.19) and integrate all intermediate endpoints over their boxes. Their widths $2\sqrt{H_j}$ cancel the respective density factors. Only the last, fixed-width endpoint window leaves a factor $H_B^{-1/2}$. Thus the mass of paths satisfying the stronger barriers and ending in $\mathcal{I}_B$ is at least

$$c^B H_B^{-1/2} \exp\left(-C \sum_j U_j^2\right) \geq L^{-C*}.$$

Indeed $B = O(\ell)$, (3.17) holds, and $\log H_B = O(\log \ell) + O(\log r_*) = o(\ell)$ for each fixed r_* . Finally all noises together change any height by at most $mL^{-10} \leq L^{-9}$. The strict barrier margins and the endpoint interval $[y + 2/5, y + 3/5]$ absorb them. On (3.12) the terminal weight is at least e^{-s} . Increasing C_* proves (3.14).

Endpoint costs and the order of constants. We give the bookkeeping behind the two cancellations just used. From regularity, $|q - \kappa H_j| \leq E_j \sqrt{H_j}$. Furthermore $\max_j E_j / \sqrt{H_j} \leq K_E \sqrt{D_0} \log \ell / r = o(1)$. Thus q/H_j lies, for all large n , in a positive compact interval determined by the original κ interval. This bound comes from the coarse environment energy; it does not require any further restriction on the fine-count upper tolerance.

The interior gap differences in (3.15) involve at most three neighboring E 's. Adjacent square-root widths have ratios between $1/\sqrt{5}$ and $\sqrt{5}$. The inequality $(t_1 + \dots + t_5)^2 \leq 5 \sum_{\nu=1}^5 t_\nu^2$, applied to (3.16) and then summed in j , consequently gives

$$\sum_j U_j^2 \leq C \left(B + \sum_j E_j^2 + r^2/H_1 + r^2/H_B \right) \leq C(1 + C_E)B + 2CD_0 \log \ell.$$

Every interior E_j^2 is counted at most three times. Also $U_j/\sqrt{H_j} = o(1)$ uniformly, including the two boundary terms $r/H_j \leq D_0 \log \ell / r$. These bounds justify use of the fixed near-mean neighborhood from Lemma 3.2 for every pair of box endpoints.

For the path inclusion, write $\tau = a_i - a_{i_{j-1}}$ and $D_i = S_i - S_{i_{j-1}} - \kappa\tau$. Since $q = \kappa H_j + D_{i_j}$, elementary cancellation gives

$$\tau - \frac{\kappa\tau + D_i}{q} H_j = \frac{\tau D_{i_j} - H_j D_i}{q}.$$

Here $0 \leq \tau \leq H_j$ and both deviations are at most $E_j \sqrt{H_j}$ in modulus. This is the error charged against the two adjacent endpoint gaps in (3.18). At the initial and final endpoints the additional gap r/s supplies the same margin by the choice of D_0 and (3.13). The final endpoint is centered at $y + 1/2$, so its gap is $r/s - 1/2$, exactly as specified above.

Tonelli's theorem and the independence of disjoint groups identify the constrained endpoint-chain mass with the iterated integral of their killed kernels. The quantitative lower bound for that integral is

$$\left(\prod_{j=1}^B c H_j^{-1/2} e^{-C U_j^2} \right) \left(\prod_{j=1}^{B-1} 2 \sqrt{H_j} \right) \frac{1}{5} = \frac{2^{B-1} c^B}{5} H_B^{-1/2} e^{-C \sum_j U_j^2}.$$

Taking logarithms leaves a loss at most $C' B + C'' \sum_j U_j^2 + \frac{1}{2} \log H_B + C'''$. The coefficient of ℓ in this bound can be fixed before g_*, r_* : $B \leq C \ell$ with a fixed coefficient, the energy is bounded as above, and $\log H_B = O(\log \ell) + O(|\log r_*| + |\log D_0|)$. Only the threshold for n changes when these fixed parameters change. Finally one can first fix any noise vector satisfying the coordinate bound, apply the deterministic margin argument, and then integrate that vector. Thus this lower bound even allows dependence among the noise coordinates; the noise vector is still sampled independently of the increment path. $\square$

Remark 3.5 (Why the exponent is logarithmic). A separate fixed-width endpoint condition in every coarse group would produce $\prod_j H_j^{-1/2}$ and a much larger logarithmic cost. The pointwise kernel estimate permits boxes of width $\sqrt{H_j}$ and cancels all intermediate factors by integration. Only $\sum E_j^2$, not $B \max E_j^2$, is charged. These two facts are essential to the precision in Theorem 1.1.

4 Fourier remainders for the marked height kernels

These estimates concern only the explicitly evaluated kernels (2.9). The coefficient-reservoir theorem is what allows them to be used under the original size and cycle constraints.

4.1 Fourier regularity and arithmetic separation

For $\operatorname{Re} z > 0$ put $\phi_z(t) = |1 - e(t)|^z$, set its value at an integer to zero, and use the real logarithm of $|1 - e(t)|$ off the integers. Use the convention $\hat{\phi}_z(j) = \int_{\mathbb{T}} \phi_z(t) e(-jt) dt$. For $s > 0$ its Fourier coefficients are

$$\hat{\phi}_s(j) = \frac{(-1)^j \Gamma(1+s)}{\Gamma(1+s/2-j) \Gamma(1+s/2+j)}. \quad (4.1)$$

The formula follows from the beta integral; alternatively the denominator-free recurrence $(s/2+j)\hat{\phi}_s(j) = -(1+s/2-j)\hat{\phi}_s(j-1)$ for $j \geq 1$, together with $\hat{\phi}_s(0) = A(s)$ and $\hat{\phi}_s(-j) = \hat{\phi}_s(j)$, determines the coefficients. Reciprocal gamma zeros are interpreted literally.

Lemma 4.1 (Uniform Fourier tail). *Fix $0 < p_- < p_+ < \infty$ and $0 < \alpha < \min(p_-, 1)$. For $p \in [p_-, p_+]$, $u \in \mathbb{R}$, and every $R > 0$,*

$$\sum_j |\hat{\phi}_{p+iu}(j)| \leq C(1+|u|)^2, \quad \sum_{|j|>R} |\hat{\phi}_{p+iu}(j)| \leq C(1+|u|)^2 R^{-\alpha}. \quad (4.2)$$

Proof. The function is absolutely continuous and its derivative is integrable, since near an integer its magnitude is bounded by $C(1+|u|)|t|^{p_- - 1}$. Split an L^1 translation difference of the derivative at distance h from the integers. Within that distance its integral is at most

$C(1 + |u|)h^{p_-}$. Away from it, integrate the second derivative bound $C(1 + |u|)^2(|t|^{p_- - 2} + 1)$ over the translation path. This gives

$$\|\phi'_{p+iu}(\cdot + h) - \phi'_{p+iu}\|_1 \leq C(1 + |u|)^2 h^\alpha.$$

The smaller exponent α absorbs the logarithm at $p_- = 1$. Use $h = 1/(2|j|)$ in the Fourier coefficient of this difference, and $\widehat{\phi}'(j) = 2\pi i j \widehat{\phi}(j)$. It follows that $|\widehat{\phi}_{p+iu}(j)| \leq C(1 + |u|)^2 |j|^{-1-\alpha}$ for $j \neq 0$. The zero coefficient is bounded by $A(p)$. Summation proves the lemma. $\square$

For a nonempty harmonic block $I(a, c) = \{j : e^a < j \leq e^c\}$ let $H = \sum_{j \in I(a, c)} 1/j$ and define

$$B_{a, c}(\mathbf{z}; \mathbf{t}) = \frac{1}{H} \sum_{j \in I(a, c)} \frac{1}{j} \prod_{v=1}^d \phi_{z_v}(j t_v), \quad d = 1, 2.$$

Summation by parts in a geometric progression yields

$$\left| \sum_{j \in I(a, c)} \frac{e(ju)}{j} \right| \leq \frac{C e^{-a}}{\|u\|_{\mathbb{T}}} \quad (u \notin \mathbb{Z}). \quad (4.3)$$

It follows from Lemma 4.1 that whenever

$$\|k_1 t_1 + \cdots + k_d t_d\|_{\mathbb{T}} \geq e^{-a+\Delta} \quad (0 \neq \mathbf{k} \in \mathbb{Z}^d, |k_v| \leq R), \quad (4.4)$$

Re $z_v \in [p_-, p_+]$, and $|\Im z_v| \leq T_0$, one has

$$\left| B_{a, c}(\mathbf{z}; \mathbf{t}) - \prod_v A(z_v) \right| \leq C(1 + T_0)^{2d} \left(R^{-\alpha} + \frac{e^{-\Delta}}{H} \right). \quad (4.5)$$

To see this, expand the absolutely convergent Fourier series. The zero vector contributes the stated product. On the box $|k_v| \leq R$ use (4.3); off it use one tail estimate and the remaining absolute Fourier norms. This also proves uniformity when the observation points move with n , subject only to (4.4).

4.2 Smoothed kernel comparison with relative-event accuracy

Lemma 4.2 (Polynomially accurate kernel comparison). *For each fixed $J > 0$ there are positive constants u_1, u_2, u_3 such that the following holds, uniformly in the compact κ interval. Put*

$$\delta = L^{-10}, \quad T_0 = L^{u_1}, \quad R = \lceil L^{u_2} \rceil, \quad \Delta = u_3 \ell. \quad (4.6)$$

*In a fine block with $q \asymp \omega \asymp \ell$, take q harmonic samples and their d log-sine sums X_v , $d = 1, 2$. Suppose (4.4) holds. Tilt their joint height measure by $e^{s \sum_v X_v}$ and normalize. Add independently to each coordinate a noise ζ which is the sum of ten uniforms on $[-\delta/10, \delta/10]$. This normalized smoothed measure differs in total variation by at most L^{-J} from d independent copies of f_s^{*q} convolved with that noise. Also*

$$B_{a, c}(s, \dots, s; \mathbf{t})^q = A(s)^{dq} \{1 + O(L^{-J})\}. \quad (4.7)$$

Proof. Choose a positive $\alpha < \min(s_{\min}/2, 1)$, and apply (4.5) to a real tilt interval containing $s/2, s$ and $3s/2$ for every possible s . The characteristic function of the tilted vector is

$$\left\{ \frac{B_{a, c}(s + it'_1, \dots, s + it'_d; \mathbf{t})}{B_{a, c}(s, \dots, s; \mathbf{t})} \right\}^q.$$

On $[-T_0, T_0]^d$ compare it with $\{\prod_v A(s + it'_v)/A(s)^d\}^q$. The normalized quantities have modulus at most one, so their q th powers differ by at most $Cq(1 + T_0)^{2d}(R^{-\alpha} + e^{-\Delta}/H)$. The real denominator is bounded away from zero for large n .

The noise transform has modulus at most $\min(1, C(\delta|t|)^{-10})$. Fourier inversion bounds the uniform density error inside the frequency box by $CT_0^d q(1 + T_0)^{2d}(R^{-\alpha} + e^{-\Delta}/H)$; outside it the bound is $C\delta^{-d}(\delta T_0)^{-9}$. For example, first choose u_1 with $9u_1 > J + 160$, then u_2, u_3 with

$$\alpha u_2 > 6u_1 + J + 30, \quad u_3 > 6u_1 + J + 30.$$

The density error is then $O(L^{-J-10})$. On $[-L^2, L^2]^d$ integration costs at most $O(L^4)$. Outside this spatial box positive tails vanish, since $X_v \leq q \log 2$. Under the tilt, a negative tail is at most

$$e^{-sL^2/3} \left\{ \frac{B_{a,c}(s/2, s, \dots, s; \mathbf{t})}{B_{a,c}(s, \dots, s; \mathbf{t})} \right\}^q \leq e^{-cL^2}.$$

The ratio is bounded by (4.5); the reference measures have the same tail bound. This proves the total variation assertion. (4.7) follows from the real version of (4.5), increasing the fixed exponents if necessary. Atoms at $-\infty$ have zero tilted mass. No total variation comparison with a continuous kernel is asserted before smoothing.

Frequency tails, normalization, and product errors. To make the frequency estimate explicit, let h_δ be the characteristic function of the sum of the ten uniforms. Direct integration of a uniform variable gives

$$h_\delta(t) = \left(\frac{\sin(\delta t/10)}{\delta t/10} \right)^{10}, \quad \int_{\mathbb{R}} |h_\delta(t)| dt \leq C\delta^{-1}.$$

The ratio at zero is interpreted as one. For $\delta T_0 \geq 1$,

$$\int_{|t| > T_0} |h_\delta(t)| dt \leq C\delta^{-1}(\delta T_0)^{-9}.$$

For $d = 1, 2$, the complement of the frequency cube is covered by the d events $|t_v| > T_0$. Integrating the product noise transform and using these two bounds gives $C\delta^{-d}(\delta T_0)^{-9}$. Both normalized characteristic functions have modulus at most one, which justifies this bound for their difference as well. With $\delta = L^{-10}$ its worst exponent, at $d = 2$, is $110 - 9u_1$. Inside the cube the exponent is at most $6u_1 + 1 - \min(\alpha u_2, u_3)$, using $q \leq L$ and $H \geq 1$. The stated choices make both bounds smaller than L^{-J-10} after absorbing fixed constants. The integration volume $O(L^4)$ then remains within the total variation budget.

The normalizer is controlled separately from this total variation estimate. If a real block normalizer is $A(s)^d(1 + e_i)$, its relative error tends to zero uniformly, and $|e_i| \leq 1/2$ for large n . The inequality $|\log(1 + e_i)| \leq 2|e_i|$ shows that raising it to the q_i th power costs at most $2q_i|e_i|$ in its logarithm. The frequency exponents can be increased once so that the desired L^{-J} bound includes this factor. On multiplying the fine block normalizers the logarithmic errors add.

For normalized probability kernels μ_i, ν_i , changing one factor at a time in their product proves

$$\left| \int F d \bigotimes_{i=1}^m \mu_i - \int F d \bigotimes_{i=1}^m \nu_i \right| \leq \sum_{i=1}^m \|\mu_i - \nu_i\|_{\text{TV}}, \quad 0 \leq F \leq 1.$$

One may take total variation here to be the supremum over tests in $[0, 1]$; the L^1 density bound also bounds this quantity. In each telescoping term the other factors are probability measures, so the same inequality applies however many coordinates the path test F examines. For the present kernels the resulting bound is mL^{-J} , with the same choice of exponents for one and two observation points.

□

4.3 An integrated bound for every arithmetic arc

For $s > 0$ define

$$b_s = \begin{cases} 0, & 0 < s \leq 2, \\ (s-1) \log 2 - \lambda(s), & s > 2. \end{cases} \quad (4.8)$$

Lemma 4.3 (Entropy gap and a block envelope). *At $s = s_\kappa$ one has $\kappa b_s < 1$, with a uniform strict gap on the compact κ interval. For every block,*

$$B_{a,c}(s; t) \leq A(s) \exp\{b_s + C/H\} \quad (t \in \mathbb{T}). \quad (4.9)$$

Proof. For an arbitrary interval of positive integers, $\sum \cos(2\pi ju)/j \geq -C$, uniformly in u . Indeed the cosines are nonnegative up to $(6\|u\|_{\mathbb{T}})^{-1}$; the remaining sum is bounded by (4.3) starting at that cutoff. If the cutoff is below one, start at one, and if u is integral the sum is positive. For $0 < s \leq 2$, all nonzero coefficients in (4.1) are nonpositive. Pairing them and applying the preceding lower bound gives $B_{a,c}(s; t) \leq A(s) + C/H$. For $s > 2$ use $\phi_s \leq 2^{s-2}\phi_2$, giving $B_{a,c}(s; t) \leq 2^{s-1} + C/H$. This proves (4.9).

The gamma duplication formula and the integral for the difference of two digamma functions give

$$\log 2 - \lambda'(s) = \int_0^\infty \frac{e^{-sx}}{1+e^x} dx, \quad 0 < s(\log 2 - \lambda'(s)) < \frac{1}{2}.$$

For example, the integral follows by subtracting the two logarithmic derivatives in the duplication formula and expanding $(1+e^x)^{-1}$, or directly integrating the convergent digamma difference. Thus for $s > 2$, (1.6) gives

$$\kappa b_s = 1 + \kappa\{s(\log 2 - \lambda'(s)) - \log 2\} < 1.$$

For $s \leq 2$ the assertion is immediate. Continuity and compactness make the gap uniform. $\square$

Choose $\eta > 0$ so small that $(\kappa + \eta)b_{s_\kappa} < 1$ uniformly. Use this value in (3.10). If $X_i(t)$ denotes the harmonic height sum in fine block i and $X_{\text{mid}} = \sum_i X_i$, then

Lemma 4.4 (Integrated middle moment). *For sufficiently large fixed r_* (as specified below), uniformly over regular count vectors,*

$$\mathbb{E} \int_{\mathbb{T}} e^{sX_{\text{mid}}(t)} dt \leq A(s)^Q \{1 + o(1)\}. \quad (4.10)$$

The expectation here is evaluation of the normalized product of harmonic kernels, not an assumption about independence in the original Ewens law.

Proof. Let $d_R(t) = \min_{1 \leq |j| \leq R} \|jt\|_{\mathbb{T}}$. For $d_R(t) \geq e^{-r+\Delta}$ every block is separated and (4.7) gives $A(s)^Q(1 + o(1))$. Partition the other points according to

$$e^{-a_k+\Delta} \leq d_R(t) < e^{-a_{k-1}+\Delta}, \quad 1 \leq k \leq m,$$

including all smaller distances in class m . That class has measure at most $CRe^{-a_{k-1}+\Delta}$. Blocks after k are separated. Using (4.9), $q_i \leq (\kappa + \eta)\omega$, and $\omega \rightarrow \infty$, there is a fixed $\rho < 1$ such that

$$\prod_{i \leq k} B_{a_{i-1}, a_i}(s; t)^{q_i} \leq A(s)^{S_k} e^{\rho(a_k - r)}.$$

The per-block Cq_i/H_i errors contribute $O(k)$, which is absorbed by the uniform strict gap since $a_k - r = k\omega$. The suffix contributes $A(s)^{Q-S_k}(1 + o(1))$; no suffix is required for $k = m$. The normalized integral over all these classes is at most

$$CL^{u_2+u_3} e^{-r+\rho\omega} \sum_{k=1}^m e^{-(1-\rho)(a_{k-1}-r)} = o(1)$$

when $r_* > u_2 + u_3 + A_0 + 2$. The geometric sum is bounded. $\square$

5 Closing the maximum bounds inside the exact coefficients

5.1 Order of constants

All choices are uniform over the fixed interval of κ . Choose η from Lemma 4.3, then A_0, K_E, C_E from Lemma 3.3. Select D_0, C_* in Lemma 3.4. Next fix $J > 2C_* + 12$ and select u_1, u_2, u_3 from Lemma 4.2. Finally take g_*, r_* so large that

$$\begin{aligned} s_{\min} g_* &> 2u_2 + u_3 + A_0 + 2C_* + 12, \\ r_* &> 64s_{\max} g_* + 1, \quad r_* > 2u_2 + u_3 + A_0 + 12. \end{aligned} \quad (5.1)$$

This is not circular: Lemma 3.4 explicitly made C_* independent of the last two constants. Throughout this section the count vector is regular, until its probability is restored at the end.

5.2 A first and relative second kernel calculation

Let the harmonic product measure at fixed counts be denoted by $\mathbb{E}_H$. Define

$$\mathcal{D}_L = \{t \in \mathbb{T} : \|jt\|_{\mathbb{T}} \geq e^{-r+\Delta}, 1 \leq |j| \leq R\}.$$

Its complement has measure at most $CR e^{-r+\Delta} = o(1)$. The following lower bound is uniform over every measurable $\mathcal{D} \subset \mathcal{D}_L$ of measure at least $1/2$. Let ζ_i be independent noises as in Lemma 4.2, put $U_\zeta = \sum_i \zeta_i$, and define a nonnegative functional of the actual harmonic samples by

$$Z(t) = \mathbb{E}_\zeta \left[e^{-sU_\zeta} \mathbf{1}_{\{(\sum_{j \leq i} (X_j(t) + \zeta_j))_{i \leq m} \in \mathcal{A}\}} \right], \quad Z_{\mathcal{D}} = \int_{\mathcal{D}} Z(t) dt. \quad (5.2)$$

The event $\mathcal{A}$ is exactly (3.12); the notation means its cumulative-height inequalities. If $Z_{\mathcal{D}} > 0$, then some actual point in $\mathcal{D}$ has middle height at least $y - m\delta$. The noises are auxiliary integrations, not alterations of the final spectral field.

For $t \in \mathcal{D}_L$, tilt each fine height kernel by $e^{sX_i(t)}$. The normalizer is $A(s)^Q(1 + o(1))$. The noise factor in (5.2) combines with the inverse tilt to give the terminal weight $e^{-s \sum_i (X_i(t) + \zeta_i)}$. After removing e^{-sy} , the test function is bounded by one on $\mathcal{A}$. Telescoping the product of the normalized smoothed measures gives error at most mL^{-J} . Since $m \leq L$ and $p_L \geq L^{-C_*}$,

$$mL^{-J} = o(p_L^2), \quad (5.3)$$

not just $o(1)$. Lemma 4.2 therefore gives uniformly in t

$$\mathbb{E}_H Z(t) = (1 + o(1)) \mu_L p_L, \quad \mu_L = e^{-sy + \lambda Q} = e^{-a_* + r}. \quad (5.4)$$

Consequently $\mathbb{E}_H Z_{\mathcal{D}} = (1 + o(1)) \text{Leb}(\mathcal{D}) \mu_L p_L$.

For a pair define

$$d_R(t, u) = \min_{\substack{(j, l) \in \mathbb{Z}^2 \setminus \{(0, 0)\} \\ |j|, |l| \leq R}} \|jt + lu\|_{\mathbb{T}}.$$

If $d_R(t, u) \geq e^{-r+\Delta}$, apply the two-point comparison in every block. The reference kernels at the two points are independent, including independent noise vectors in the two copies of (5.2). Thus

$$\mathbb{E}_H [Z(t)Z(u)] = (1 + o(1)) \mu_L^2 p_L^2 \quad (5.5)$$

uniformly over these far pairs. Equation (5.3) is what makes this relative statement valid.

The bounded tests after the change of measure. Let $\mathcal{N}_t$ be the product of the one-point block normalizers and let $\tilde{\mathbb{P}}_t$ be the corresponding product tilted law, before adjoining the noises. Set $\tilde{W}_i = \sum_{j \leq i} (X_j(t) + \zeta_j)$. The exact identity, with no asymptotic comparison yet, is

$$\mathbb{E}_H Z(t) = e^{-sy} \mathcal{N}_t \tilde{\mathbb{E}}_{t,\zeta} [e^{-s(\tilde{W}_m - y)} \mathbf{1}_{\mathcal{A}}].$$

Indeed the inverse tilt contributes $e^{-s \sum_i X_i(t)}$, and the factor e^{-sU_ζ} in $Z(t)$ supplies the missing noise sum. On $\mathcal{A}$, $0 \leq \tilde{W}_m - y \leq 1$, so the bracketed test lies in $[0, 1]$. Replacing its smoothed tilted product law by the reference law costs at most mL^{-J} . The reference expectation is the same p_L for every t . The remaining prefactor is $e^{-sy} A(s)^Q (1 + o(1)) = \mu_L (1 + o(1))$.

For a far pair use the joint tilt $e^{s \sum_i (X_i(t) + X_i(u))}$. In the product $Z(t)Z(u)$ the two noise integrals are independent copies. After inverse tilting the test is the product of two tests in $[0, 1]$. The reference one-point paths are independent under the reference two-point law, so its expectation is exactly p_L^2 . It is this reference independence, supplied by two-point Fourier separation, that is used in (5.5). Since

$$\frac{mL^{-J}}{p_L} \leq L^{1-J+C_*}, \quad \frac{mL^{-J}}{p_L^2} \leq L^{1-J+2C_*},$$

both errors tend to zero under the one chosen inequality $J > 2C_* + 12$. Relative normalizer errors also tend to zero. No lower bound on the unnormalized rare probability $\mu_L p_L$ is used for the smoothing step; its exponential scale has already been extracted exactly.

For non-far pairs let class k be

$$e^{-a_k + \Delta} \leq d_R(t, u) < e^{-a_{k-1} + \Delta}, \quad 1 \leq k \leq m,$$

including every smaller distance in class m . The measure of that class in $\mathbb{T}^2$ is at most

$$CR^2 e^{-a_{k-1} + \Delta}. \quad (5.6)$$

Every nonzero integer linear map from $\mathbb{T}^2$ to $\mathbb{T}$ preserves uniform measure, so a union bound over the $O(R^2)$ maps proves this estimate. Blocks after k are two-point separated.

On both noisy path events, the actual terminal sums are at least $y - m\delta$, whereas the prefix at u through block k is at most $B_k + m\delta$. Applying exponential Markov to the terminals and using that prefix bound yields

$$\begin{aligned} \mathbb{E}_H [Z(t)Z(u)] &\leq e^{-2sy + sB_k + O(sm\delta)} \mathbb{E}_H e^{s \sum_{i \leq k} X_i(t)} \mathbb{E}_H e^{s \sum_{i > k} (X_i(t) + X_i(u))} \\ &\leq \exp\{-2sy + sB_k + \lambda S_k + 2\lambda(Q - S_k) + o(1)\}. \end{aligned} \quad (5.7)$$

The two factors in the first line use disjoint blocks, which are product kernels under $\mathbb{E}_H$. The prefix uses only one-point separation of $t \in \mathcal{D}_L$; it does not assume independence of the two unseparated prefixes. The suffix uses two-point separation after k . For $k = m$ it is exactly one. The bounded noise weights cause only the displayed $O(sm\delta)$ error. Since $sB_k = a_k + \lambda S_k - sG$, division by $\mu_L^2 p_L^2$ gives

$$\frac{\mathbb{E}_H [Z(t)Z(u)]}{\mu_L^2 p_L^2} \leq C e^{a_k} L^{-s_{\min} g_* + 2C_*}. \quad (5.8)$$

Multiply by (5.6) and sum over $m \leq L$ classes. Because $a_k - a_{k-1} = \omega \leq A_0 \ell$, the normalized near contribution is at most

$$CL^{1+2u_2+u_3+A_0-s_{\min} g_*+2C_*} = o(1). \quad (5.9)$$

Thus the exponential correlation bound and the arithmetic area cancel at the correct scale. The polynomial gap G beats all remaining errors.

The total measure of non-far pairs is $O(R^2 e^{-r+\Delta}) = o(1)$. Equations (5.4), (5.5), and (5.9) show

$$\frac{\mathbb{E}_H Z_{\mathcal{D}}^2}{(\mathbb{E}_H Z_{\mathcal{D}})^2} \longrightarrow 1$$

uniformly over the stated environments and all choices of $\mathcal{D}$. Chebyshev's inequality proves

$$\mathbb{P}_H \left(\sup_{t \in \mathcal{D}} X_{\text{mid}}(t) < \frac{a_* - r + \lambda Q}{s} - m\delta \right) = o(1). \quad (5.10)$$

All integrals defining these functionals are justified by nonnegativity; $Z(t) \leq e^{sm\delta}$. Singularities of the logarithm at roots have zero tilted mass and cannot create a high point.

The last near class and the uniform second-moment conclusion. The inequality for near pairs can be verified before taking any expectation. For fixed noises write T_t, T_u for the two actual terminal heights, P_t, P_u for the actual prefixes through block k , and R_t, R_u for the two actual suffix heights after block k . On the two path events, $T_t, T_u \geq y - m\delta$ and $P_u \leq B_k + m\delta$. After dropping the indicators and bounding the two noise weights, the integrand is bounded by

$$\exp\{-2sy + sB_k + Csm\delta\} \exp\{sP_t + s(R_t + R_u)\}.$$

This follows by multiplying the two terminal exponential Markov bounds and then using $T_u = P_u + R_u$. Only the t -prefix remains random in the bound; the u -prefix has been bounded by its barrier. Products across prefix and suffix blocks factor under the actual harmonic product law. At $k = m$ both suffixes are empty, their exponential factor and expectation are one, and the same bound is valid. Thus pairs with arbitrarily small or zero arithmetic separation are included in the last class.

For the area estimate, a nonzero integer vector (j, l) defines a surjective homomorphism $(t, u) \mapsto jt + lu$ of tori, so the inverse image of a distance- ϵ interval has Haar measure at most 2ϵ . Summing over at most $(2R+1)^2 - 1$ vectors proves (5.6), also if some vectors have common divisors or one coordinate is zero. The cancellation in the normalized moment is exact:

$$-2sy + sB_k + \lambda S_k + 2\lambda(Q - S_k) - \log(\mu_L^2) = a_k - sG.$$

Hence its only nonpolynomial factor, e^{a_k} , is canceled by $e^{-a_{k-1}}$ from area, leaving $e^\omega \leq L^{A_0}$. The finite last class uses the same area bound and requires no separated suffix.

For completeness, the passage from these bounds to probability is uniform in the measurable set $\mathcal{D}$. Put $d_{\mathcal{D}} = \text{Leb}(\mathcal{D}) \geq 1/2$ and $a_L = d_{\mathcal{D}} \mu_L p_L > 0$. The first moment is $a_L(1 + o(1))$. On far pairs the second-moment integrand is at most $(1 + o(1))\mu_L^2 p_L^2$, whose integral over $\mathcal{D}^2$ is at most $(1 + o(1))a_L^2$. The entire near integral is $o(\mu_L^2 p_L^2) = o(a_L^2)$, since $d_{\mathcal{D}}^{-2} \leq 4$. Thus

$$\mathbb{P}_H(Z_{\mathcal{D}} = 0) \leq \frac{\text{Var}_H(Z_{\mathcal{D}})}{(\mathbb{E}_H Z_{\mathcal{D}})^2} = \frac{\mathbb{E}_H Z_{\mathcal{D}}^2}{(\mathbb{E}_H Z_{\mathcal{D}})^2} - 1 = o(1).$$

For example, if the first moment is at least $(1 - \epsilon)a_L$ and the second is at most $(1 + \epsilon)a_L^2$, with $0 < \epsilon < 1$, the last bound is at most $(1 + \epsilon)/(1 - \epsilon)^2 - 1$. All bounds depend on $\mathcal{D}$ only through its lower mass bound. Also $Z_{\mathcal{D}} > 0$ entails a point and a noise realization with terminal height at least y ; the actual height there is at least $y - m\delta$. This completes the lower estimate with its required uniformity over every such $\mathcal{D}$.

5.3 The upper kernel calculation

For a nonzero polynomial P of degree $d \geq 1$,

$$\sup_{|z| \leq 1} |P'(z)| \leq ed \|P\|_{\infty}, \quad \int_{\mathbb{T}} |P(e(t))|^s dt \geq \frac{c_s}{d} \|P\|_{\infty}^s. \quad (5.11)$$

For the first estimate, the maximum principle applied to $P(z)/z^d$ outside the disk gives $|P(z)| \leq |z|^d \|P\|_\infty$. Cauchy's estimate on a circle of radius $1/d$ about a point in the unit disk then gives the derivative bound. An arc of length c/d around a boundary maximizer has modulus at least half the maximum, proving the second estimate. The constants may be uniform for s in its compact interval. The middle polynomial has degree at most $Qb \leq CLb$ deterministically. Hence Lemma 4.4 and Markov's inequality imply, for a sufficiently large fixed C_1 ,

$$\mathbb{P}_H \left(\max_t X_{\text{mid}}(t) > \frac{a_* + \lambda Q}{s} + C_1 \ell \right) \leq CL^{1-s_{\min} C_1} = o(1). \quad (5.12)$$

Indeed this event forces the integral in (4.10) to be at least $cA(s)^Q L^{sC_1-1}$. No spatial discretization assumption is needed.

5.4 Restoring low cycles before the coefficient is completed

Let $q_0 \leq Cr$ and write X_0 for the low field. For every deterministic low configuration,

$$\int_{\mathbb{T}} (-X_0(t))_+ dt \leq c_V q_0, \quad c_V = \int_{\mathbb{T}} (-\log |1 - e(t)|)_+ dt < \infty.$$

Choose fixed C_2 so large that $\{X_0 < -C_2 r\}$ has measure at most $1/8$. For large n its complement intersected with $\mathcal{D}_L$ is a measurable set $\mathcal{D}$ of measure at least $1/2$. Under the harmonic product measure it depends only on low frequencies, which are independent of the middle frequencies. The bound (5.10) is uniform over every such measurable set, so it applies conditionally to this choice. Together with $X_0(t) \leq q_0 \log 2$, (5.12), and $q_0 = O(r)$, it follows that

$$\mathbb{P}_H \left(\left| \log \left\| \prod_{j \leq b} (1 - z^j)^{C_j} \right\|_\infty - \frac{a_* + \lambda(k-l)}{s} \right| > C_3 \ell \right) = o(1). \quad (5.13)$$

Here the counts are fixed, $q_0 + Q = k - l$, $r = r_* \ell$, and the constants remain uniform on all regular count vectors with the bounds above.

Now apply Theorem 2.3 to the full short-cycle configuration, including the low part, and to the indicator of the event in (5.13). Its degree is at most kb and its coefficients are positive. The actual conditional failure probability, given the same count vector, is at most $(1 + O(\varepsilon_n))$ times its harmonic-product failure probability. This is a single comparison after all projections; there is no factor $(1 + \varepsilon_n)^m$ and no subtraction of an error bigger than the high-point probability. Finally Lemma 3.3 shows that these count vectors have actual conditional probability tending to one. Therefore (5.13) holds under $\mathbb{P}_{n,k}$ as well, uniformly in k , with its random $l = L_* \leq C\ell$.

Why the random low-frequency set is admissible. The deterministic estimate for X_0 uses $(-\sum_v x_v)_+ \leq \sum_v (-x_v)_+$ and invariance of Haar measure under $t \mapsto jt$, for each positive integer frequency j . Consequently

$$\text{Leb}\{t : X_0(t) < -C_2 r\} \leq \frac{c_V q_0}{C_2 r} \leq \frac{1}{8}$$

after one fixed choice of C_2 . Roots form a finite set and do not affect this integral or the measure bound. Together with $\text{Leb}(\mathcal{D}_L^c) = o(1)$ this gives the claimed measurable $\mathcal{D}$ with mass at least one half. For every realized low configuration apply the just-proved uniform lower estimate to that particular set. The middle harmonic sample law is unchanged by this conditioning. Integrating over low configurations therefore preserves the same upper bound on its failure probability.

The centering adjustment at this step is explicit:

$$\frac{a_* + \lambda(q_0 + Q)}{s} - \left(\frac{a_* - r + \lambda Q}{s} - C_2 r - m\delta \right) = \frac{r + \lambda q_0}{s} + C_2 r + m\delta = O(\ell).$$

For the upper estimate one adds at most $q_0 \log 2$ to the middle maximum. The compact bounds on s and $\lambda(s)$, $q_0 \leq Cr$, and the fixed value of r_* make these constants uniform.

Finally, let $\mathcal{R}_{n,k}$ denote the set of count vectors obeying all regularity and reservoir-count bounds. Its actual conditional probability tends uniformly to one. On this set the normalized order estimate bounds the conditional short-field failure by a common quantity $\frac{1+\varepsilon_n}{1-\varepsilon_n} o(1)$, independent of the individual count vector. Summing these conditional probabilities bounds the unconditional failure by

$$\mathbb{P}_{n,k}(\mathcal{R}_{n,k}^c) + \frac{1+\varepsilon_n}{1-\varepsilon_n} o(1) = o(1).$$

Only count vectors of positive actual mass enter this sum. The positive transfer also supplies positivity of the denominator whenever the reference mass is positive in the reservoir range.

5.5 Long-cycle insertion with an exact conditional moment

Lemma 5.1 (Insertion stability). *Let P be nonzero of degree $d \geq 1$ and $Q(z) = \prod_v (1 - z^{j_v})$, where all j_v are positive integers. Then*

$$-\log 2 - 2ed \sum_v j_v^{-1} \leq \log \|PQ\|_\infty - \log \|P\|_\infty \leq (\#v) \log 2. \quad (5.14)$$

For $d = 0$ the lower bound can be replaced by zero.

Proof. The upper bound is pointwise. At a maximizer z_* of P on the unit circle take $\rho = 1 - (2ed)^{-1}$. Equation (5.11) gives $|P(\rho z_*)| \geq \|P\|_\infty / 2$. Also

$$|Q(\rho z_*)| \geq \prod_v (1 - \rho^{j_v}), \quad -\log(1 - e^{-x}) \leq (e^x - 1)^{-1} \leq x^{-1} \quad (x > 0).$$

Since $\rho^j \leq e^{-j/(2ed)}$, the claimed lower bound follows from the maximum modulus principle for PQ . When P is constant use $Q(0) = 1$. $\square$

Lemma 5.2 (Conditional cross-mass bound). *For k/L in the fixed compact interval and $b \leq n/4$,*

$$\mathbb{E}_{n,k} \left[\left(\sum_{j \leq b} j C_j \right) \left(\sum_{l > b} \frac{C_l}{l} \right) \right] \leq C \quad (5.15)$$

for all sufficiently large n , uniformly in b, k .

Proof. Deleting one j -cycle and one l -cycle in (1.4), where $j \neq l$, gives the exact identity

$$\mathbb{E}_{n,k}(C_j C_l) = \frac{1}{jl} \frac{a_{n-j-l,k-2}}{a_{n,k}}.$$

Thus the expectation in (5.15) equals

$$\sum_{\substack{j \leq b < l \\ j+l \leq n}} \frac{1}{l^2} \frac{a_{n-j-l,k-2}}{a_{n,k}}. \quad (5.16)$$

If $l \leq n/2$, then $n - j - l \geq n/4$. Formula (2.4), applied at sizes in $[n/4, n]$, bounds the coefficient ratio by a constant uniformly: $\log(n - j - l) = L + O(1)$ and $k(k-1)/L^2$ is bounded. This part of the sum is at most $Cb \sum_{l > b} l^{-2} \leq C$.

For $l > n/2$, use $l^{-2} \leq 4/n^2$ and the exact cumulative identity

$$\sum_{r=0}^n a_{r,k-2} = [u^{k-2}] h_n(u+1) = n a_{n,k-1} + a_{n,k-2}. \quad (5.17)$$

It follows from $h_n(u+1) = (n+u)h_n(u)/u$. The ratios of the two coefficients on the right to $a_{n,k}$ are bounded by (2.4). Hence this part of (5.16) is at most Cb/n . This proves the assertion. The relevant k exceed two for large n , so all displayed coefficients are well defined with the conventions in (1.3). $\square$

Completion of Theorem 1.1. Under $\mathbb{P}_{n,k}$, (5.13) holds with probability tending to one, uniformly, and $L_* \leq C\ell$. For the upper bound multiply back the L_* long factors, each of supremum at most two. Since $L - a_* = 4\ell + o(1)$, this changes the comparison with $(L + \lambda k)/s$ by at most a fixed multiple of ℓ .

For the lower bound apply Lemma 5.1 with $d = \sum_{j \leq b} jC_j$. By Lemma 5.2 and Markov's inequality, $2ed \sum_{l > b} C_l/l \leq \ell$ with probability tending to one uniformly. Thus reinsertion decreases the maximum by at most $\ell + \log 2$ on that event. The difference between the two centers is

$$\frac{L + \lambda k}{s} - \frac{a_* + \lambda(k - L_*)}{s} = \frac{L - a_* + \lambda L_*}{s} = O(\ell).$$

Combining with (5.13) proves the lower half of (1.8), and hence the theorem. No lower-bound estimate for a Poisson spectral maximum was used in this completion.

A fixed localization constant before the error tolerance. Here is an explicit final choice to keep track of the quantifiers. Let C_{sh} be the fixed constant in the short-field estimate, and let C_{long} be fixed so that $L_* \leq C_{\text{long}}\ell$ with uniformly high probability. Write $\lambda_{\max} = \max_{\kappa} \lambda(s_{\kappa})$. The identity $\lambda(0) = \lambda'(0) = 0$ and $\lambda'' > 0$ gives $\lambda_{\max} > 0$. For all sufficiently large n , $0 \leq L - a_* \leq 5\ell$, so on this count event the difference of centers is nonnegative and at most $C_{\text{ctr}}\ell$, where

$$C_{\text{ctr}} = \frac{5 + \lambda_{\max} C_{\text{long}}}{s_{\min}}, \quad C = C_{\text{sh}} + C_{\text{ctr}} + C_{\text{long}} \log 2 + 1 + \log 2.$$

Increase the threshold for n so that $\ell \geq 1$. On the insertion event $2ed \sum_{l > b} C_l/l \leq \ell$, the lower loss is at most $(1 + \log 2)\ell$; the upper loss is at most $C_{\text{long}}\ell \log 2$. Both sides of the desired inequality then follow with the displayed C . If $d = 0$, the constant-polynomial version of insertion gives an even smaller lower loss.

The complement of this intersection has probability at most the sum of the short-field failure, the long-count failure, and

$$\mathbb{P}_{n,k} \left(2ed \sum_{l > b} C_l/l > \ell \right) \leq \frac{2e}{\ell} \mathbb{E}_{n,k} \left[d \sum_{l > b} C_l/l \right] \leq C_{\text{ins}}/\ell.$$

Each tends uniformly to zero. In particular, for any $\epsilon > 0$ one subsequently chooses a threshold making these three terms at most $\epsilon/3$ each. The constants defining C have already been fixed. The integer window in (1.8) is nonempty for large n because its length tends to infinity, and all its integers lie between 1 and n . For each such integer k , a permutation with one cycle of length $n - k + 1$ and $k - 1$ fixed points shows that $a_{n,k} > 0$. Thus the conditional laws and the supremum in the theorem have their literal meaning. $\square$

6 Consequences and the limits of the conclusion

For fixed $\theta > 0$ let K_n have its ordinary Ewens law. Its exact probability generating function is $h_n(\theta u)/h_n(\theta)$. The gamma-ratio estimate gives

$$\frac{K_n - \theta \log n}{\sqrt{\theta \log n}} \implies G, \quad G \sim \mathcal{N}(0, 1),$$

and $K_n/\log n \rightarrow \theta$ in probability. For example the characteristic function follows by putting $u = e^{it/\sqrt{\theta \log n}}$ in that ratio; the exponent tends to $-t^2/2$, and the gamma prefactor tends to one. Differentiation of (1.6) gives

$$v'(\kappa) = \frac{\lambda(s_\kappa)}{s_\kappa}, \quad v''(\kappa) = -\frac{1}{\kappa^3 s_\kappa^3 \lambda''(s_\kappa)}. \quad (6.1)$$

Uniform exact-cycle localization and a Taylor expansion therefore imply

Corollary 6.1 (Random centering in the ordinary Ewens law). *For every fixed $\theta > 0$, with $a_\theta = \lambda(s_\theta)/s_\theta$,*

$$M_n = v(\theta) \log n + a_\theta(K_n - \theta \log n) + O_{\mathbb{P}}(\log \log n).$$

In fact a fixed-constant $\log \log n$ bound holds with probability tending to one. Jointly,

$$\left(\frac{K_n - \theta \log n}{\sqrt{\theta \log n}}, \frac{M_n - v(\theta) \log n}{a_\theta \sqrt{\theta \log n}} \right) \Rightarrow (G, G).$$

Proof. Use Theorem 1.1 on a compact interval containing θ in its interior. The quadratic Taylor remainder in $Lv(K_n/L)$ is $O_{\mathbb{P}}(1)$, since $K_n - \theta L = O_{\mathbb{P}}(\sqrt{L})$. This is smaller than $\ell \rightarrow \infty$; the stated fixed-constant bound and the joint limit follow.

The Taylor remainder and the common Gaussian coordinate. To detail the deterministic expansion, implicit differentiation in (1.6) yields

$$\frac{ds_\kappa}{d\kappa} = -\frac{1}{\kappa^2 s_\kappa \lambda''(s_\kappa)}.$$

Substitution into $v(\kappa) = \kappa \lambda'(s_\kappa)$ gives the two derivatives in (6.1). In particular v'' is bounded on a sufficiently small fixed compact neighborhood of θ . Since $\lambda(s) > 0$ for $s > 0$, the denominator $a_\theta \sqrt{\theta L}$ in the corollary is positive. For k/L in that neighborhood Taylor's formula gives the deterministic bound

$$|Lv(k/L) - Lv(\theta) - a_\theta(k - \theta L)| \leq C_\theta \frac{(k - \theta L)^2}{L}.$$

The standardized cycle counts are tight by their CLT, so the right side, at $k = K_n$, is $O_{\mathbb{P}}(1)$. The probability of leaving the neighborhood tends to zero by the count law of large numbers.

Theorem 1.1, averaged over K_n inside this neighborhood, gives a fixed C_0 for which $\mathbb{P}(|M_n - Lv(K_n/L)| > C_0 \ell) \rightarrow 0$. For any family $R_n = O_{\mathbb{P}}(1)$ and any deterministic $\ell \rightarrow \infty$, $\mathbb{P}(|R_n| > \ell) \rightarrow 0$: given an error tolerance, first choose a uniform tightness bound and then let ℓ exceed it. Applying this observation to the Taylor remainder proves the corollary's fixed-constant assertion with $C_0 + 1$.

Set $Y_n = (K_n - \theta L)/\sqrt{\theta L}$ and let V_n denote the second coordinate in the corollary. The bound just proved implies $V_n - Y_n \rightarrow 0$ in probability, since $\ell/\sqrt{L} \rightarrow 0$. The continuous map $x \mapsto (x, x)$ sends the count CLT to $(Y_n, Y_n) \Rightarrow (G, G)$; the vanishing difference $(0, V_n - Y_n)$ then gives the displayed joint weak limit. This is a statement about joint distributions. It does not require, or assert, convergence of second moments or a limit for the Pearson correlation. $\square$

The original two-matching encoding. If a and b are fixed-point-free involutions on $2n$ labels, with a fixed and $b = gag^{-1}$ uniformly conjugated, the alternating component counts C_j have the Ewens(1/2) law in the encoding of [1]. On an alternating component of size $2j$, the product ab consists of two j -cycles. Hence

$$\det(I - zP_{ab}) = \prod_j (1 - z^j)^{2C_j}.$$

If the number of alternating components is fixed to k , Theorem 1.1 applies directly: the logarithmic maximum is within $C \log \log n$ of $2m_{n,k}$, uniformly for $k/\log n$ in a positive compact interval. This is an exact application of the encoding, not an additional extremal theorem proved by a different approximation.

Counting and identifying the actual matching model. For completeness, the encoding can be described directly. Relabel the fixed matching a so that its edges are the pairs $(i, 0), (i, 1)$, $1 \leq i \leq n$. For another matching b , the graph with edges of both matchings has even alternating components. A common edge $a = b$ is a component on two vertices; it is included. There are exactly two binary colorings of each component in which both kinds of edges join opposite colors. Consequently a matching with k components admits 2^k such colorings.

Given a coloring, denote by $\text{out}(i)$ and $\text{in}(i)$ the two vertices of the i th fixed edge, according to their colors. The matching b sends $\text{out}(i)$ to $\text{in}(\pi(i))$ for a unique permutation π of the n fixed edges. Conversely, a choice of the n orientations and a permutation π specifies all these b -edges and the coloring. The alternating components correspond exactly to the cycles of π , with twice their vertex counts. The number of permutations with cycle profile c is $n! \prod_j (j^{c_j} c_j!)^{-1}$. Counting the colored matchings in these two ways therefore gives

$$\#\{b : \text{component profile } c\} = 2^{n-k} n! \prod_j \frac{j^{-c_j}}{c_j!}, \quad \sum_j j c_j = n, \quad \sum_j c_j = k.$$

Dividing these weights by their sum identifies the component profile as Ewens(1/2), and conditioning on its component count cancels the factor $(1/2)^k$. The normalization agrees with $\#\{b\} = (2n-1)!! = 2^n n! h_n(1/2)$. Uniform conjugation indeed produces the uniform matching: all fixed-point-free involutions are conjugate, and every fiber of $g \mapsto gag^{-1}$ is a coset of the same centralizer. Relabeling back to any fixed a preserves the component profile.

The same description determines the permutation ab . On the out-vertices it acts as π and on the in-vertices as π^{-1} . A component of size $2j$ therefore contributes two j -cycles, including two fixed points when $j = 1$. A j -cycle contributes $1 - z^j$ to $\det(I - zP)$, so the determinant formula above holds for the actual product permutation. Writing $P_c(z) = \prod_j (1 - z^j)^{c_j}$, one has $\|P_c^2\|_\infty = \|P_c\|_\infty^2$ and therefore the actual logarithmic maximum equals $2 \log \|P_c\|_\infty$. Its conditioning variable is the number k of graph components, whereas the permutation ab itself has $2k$ cycles. Applying Theorem 1.1 to the size- n profile and multiplying its error bound by two proves exactly the asserted center $2m_{n,k}$.

What is and is not being asserted. The proof above closes the conditional high-probability lower bound through a coefficient comparison valid for the entire path restriction. It also supplies the matching upper localization. It does not identify $M_n - m_{n,K_n}$ after subtracting a particular multiple of $\log \log n$, nor its distribution on a smaller scale, nor an extremal point process. A previous fixed-Diophantine moment theorem and a microscopic collision kernel are not inputs here: reorganizing the generating function into a reservoir and a positive constrained polynomial avoids needing a uniform extension of those theorems over all moving singularity clusters.

The proof still uses a final second-moment inequality and elementary walk estimates. Its coefficient content is not a claim that probability can be eliminated from a probabilistic problem. The methodological point is that exact conditioning and all growing-dimensional path restrictions are handled by proved coefficient and convolution estimates, rather than being removed by an absolute-error approximation or inserted as a hypothesis.

Tools and formal verification

The mathematical development of this paper was carried out through iterative work with AI assistants, which also translated the argument into Lean 4 and ran the verification. The formalization is over the literal finite permutation model: it covers the conditional law (1.4), the characteristic polynomial as the determinant of the actual permutation matrix, Theorem 1.1, Corollary 6.1 and the two-matching application of Section 6. It was checked with `trust=0` and `debug.skipKernelTC=false`, contains no unproved placeholders, and uses only the foundational axioms `propext`, `Classical.choice` and `Quot.sound`. The Lean sources, the locked dependencies and the verification script are available at

<https://github.com/Lzp88/conditional-spectral-extremes>.

Kernel checking establishes the formal statements, not their equivalence with the prose above, and bears on neither novelty nor priority. The author is responsible for the mathematical claims.

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